CPE334 Lab 4 Final Submission Package

Course: CPE 334 Software Engineering — Lab 4 (Sprint 4)

Student Name: Pilaiwan Churdchu • **Student ID:** 67070503492 • **GitHub:** @Pilaiwan3492

Peer Reviewer: Aphichaya Klinhual • **Student ID:** 67070503447 • **GitHub:** @Apichaya251400

Author Repository: https://github.com/Pilaiwan3492/TokTickIT

Partner Repository: https://github.com/Apichaya251400/TokTickIT

Target Release: `lab4-staging` → `main` • **Release PR:** #91

Answer Part 1: Git Use with Engineering Workflow

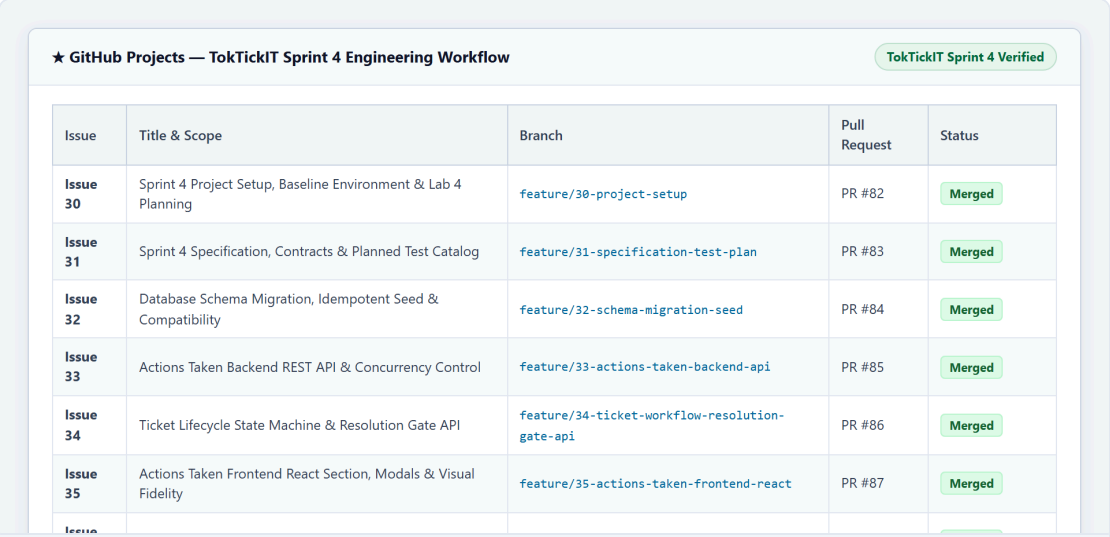
Demonstrates full Git/GitHub engineering workflow, peer-review traceability, and clean repository structure for Lab 4.

1.1 Repository / Project / PRs

All Lab 4 deliverables were coordinated and executed across dedicated feature branches, reviewed by peer reviewer @Apichaya251400, and integrated into `lab4-staging` prior to final release.

- **GitHub Repository URL:** https://github.com/Pilaiwan3492/TokTickIT
- **GitHub Project Board URL:** https://github.com/users/Pilaiwan3492/projects/1
- **GitHub Issues:**
  - Issue 30: https://github.com/Pilaiwan3492/TokTickIT/issues/72
  - Issue 31: https://github.com/Pilaiwan3492/TokTickIT/issues/73
  - Issue 32: https://github.com/Pilaiwan3492/TokTickIT/issues/74
  - Issue 33: https://github.com/Pilaiwan3492/TokTickIT/issues/75
  - Issue 34: https://github.com/Pilaiwan3492/TokTickIT/issues/76
  - Issue 35: https://github.com/Pilaiwan3492/TokTickIT/issues/77
  - Issue 36: https://github.com/Pilaiwan3492/TokTickIT/issues/78
  - Issue 37: https://github.com/Pilaiwan3492/TokTickIT/issues/79
  - Issue 38: https://github.com/Pilaiwan3492/TokTickIT/issues/80
  - Issue 39: https://github.com/Pilaiwan3492/TokTickIT/issues/81
- **GitHub Pull Requests:**
  - PR #30 (PR #82): https://github.com/Pilaiwan3492/TokTickIT/pull/82
  - PR #31 (PR #83): https://github.com/Pilaiwan3492/TokTickIT/pull/83

- PR #32 (PR #84): https://github.com/Pilaiwan3492/TokTickIT/pull/84
- PR #33 (PR #85): https://github.com/Pilaiwan3492/TokTickIT/pull/85
- PR #34 (PR #86): https://github.com/Pilaiwan3492/TokTickIT/pull/86
- PR #35 (PR #87): https://github.com/Pilaiwan3492/TokTickIT/pull/87
- PR #36 (PR #88): https://github.com/Pilaiwan3492/TokTickIT/pull/88
- PR #37 (PR #89): https://github.com/Pilaiwan3492/TokTickIT/pull/89
- PR #38 (PR #90): https://github.com/Pilaiwan3492/TokTickIT/pull/90
- PR #39 (PR #91): https://github.com/Pilaiwan3492/TokTickIT/pull/91 (Open — Pending Review)



★ GitHub Projects — TokTickIT Sprint 4 Engineering Workflow

TokTickIT Sprint 4 Verified

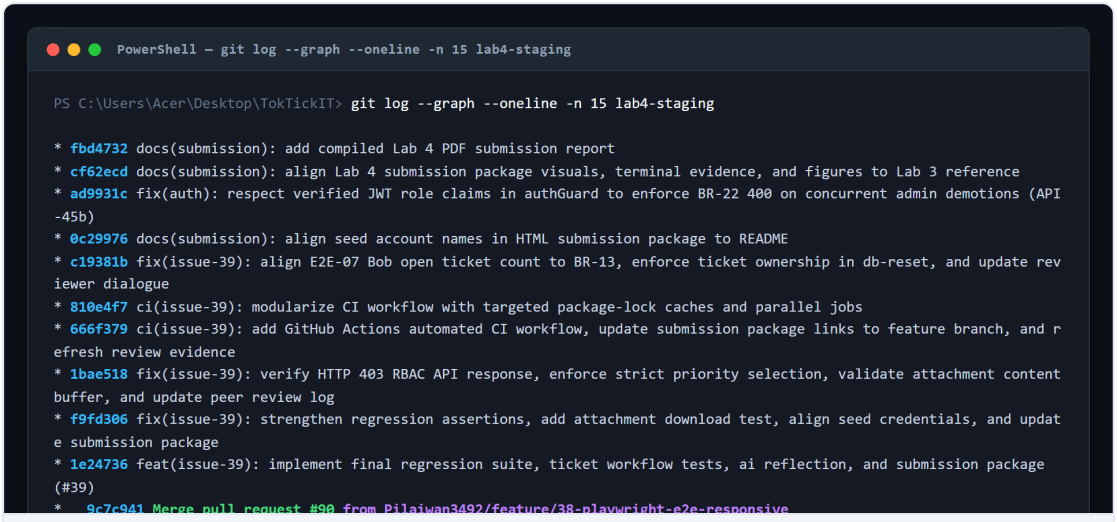
Issue	Title & Scope	Branch	Pull Request	Status
Issue 30	Sprint 4 Project Setup, Baseline Environment & Lab 4 Planning	feature/30-project-setup	PR #82	Merged
Issue 31	Sprint 4 Specification, Contracts & Planned Test Catalog	feature/31-specification-test-plan	PR #83	Merged
Issue 32	Database Schema Migration, Idempotent Seed & Compatibility	feature/32-schema-migration-seed	PR #84	Merged
Issue 33	Actions Taken Backend REST API & Concurrency Control	feature/33-actions-taken-backend-api	PR #85	Merged
Issue 34	Ticket Lifecycle State Machine & Resolution Gate API	feature/34-ticket-workflow-resolution-gate-api	PR #86	Merged
Issue 35	Actions Taken Frontend React Section, Modals & Visual Fidelity	feature/35-actions-taken-frontend-react	PR #87	Merged

Figure: Lab 4 GitHub Repository, Pull Requests, and Project Board workflow

1.2 Commit History

Clean commit history showing non-linear branch merges, issue-specific commit messages, and collaborative peer integration:

```
git log --graph --oneline -n 15 lab4-staging
```



PowerShell — git log --graph --oneline -n 15 lab4-staging

```
PS C:\Users\Acer\Desktop\TokTickIT> git log --graph --oneline -n 15 lab4-staging

* fbd4732 docs(submission): add compiled Lab 4 PDF submission report
* cf62ecd docs(submission): align Lab 4 submission package visuals, terminal evidence, and figures to Lab 3 reference
* ad9931c fix(auth): respect verified JWT role claims in authGuard to enforce BR-22 400 on concurrent admin demotions (API
-45b)
* 0c29976 docs(submission): align seed account names in HTML submission package to README
* c19381b fix(issue-39): align E2E-07 Bob open ticket count to BR-13, enforce ticket ownership in db-reset, and update rev
iewer dialogue
* 810e4f7 ci(issue-39): modularize CI workflow with targeted package-lock caches and parallel jobs
* 666f379 ci(issue-39): add GitHub Actions automated CI workflow, update submission package links to feature branch, and r
efresh review evidence
* 1bae518 fix(issue-39): verify HTTP 403 RBAC API response, enforce strict priority selection, validate attachment content
buffer, and update peer review log
* f9fd306 fix(issue-39): strengthen regression assertions, add attachment download test, align seed credentials, and updat
e submission package
* 1e24736 feat(issue-39): implement final regression suite, ticket workflow tests, ai reflection, and submission package
(#39)
*   9c7c941 Merge pull request #90 from Pilaiwan3492/feature/38-playwright-e2e-responsive
```

Figure: Terminal Git Graph commit history for Sprint 4 on lab4-staging

1.3 Project Board

All 10 Lab 4 Issues (Issues #30 through #39 / GitHub #72 through #81) have been executed and moved to the **Done** column on the Kanban board:

> **Distinction Between Issue #39 and PR #91:** Issue #39 deliverables and implementation are complete (**Done**), while PR #91 is published and awaiting formal peer review submission and approval by @Apichaya251400 (**Pending Review / Open — Not Merged**).

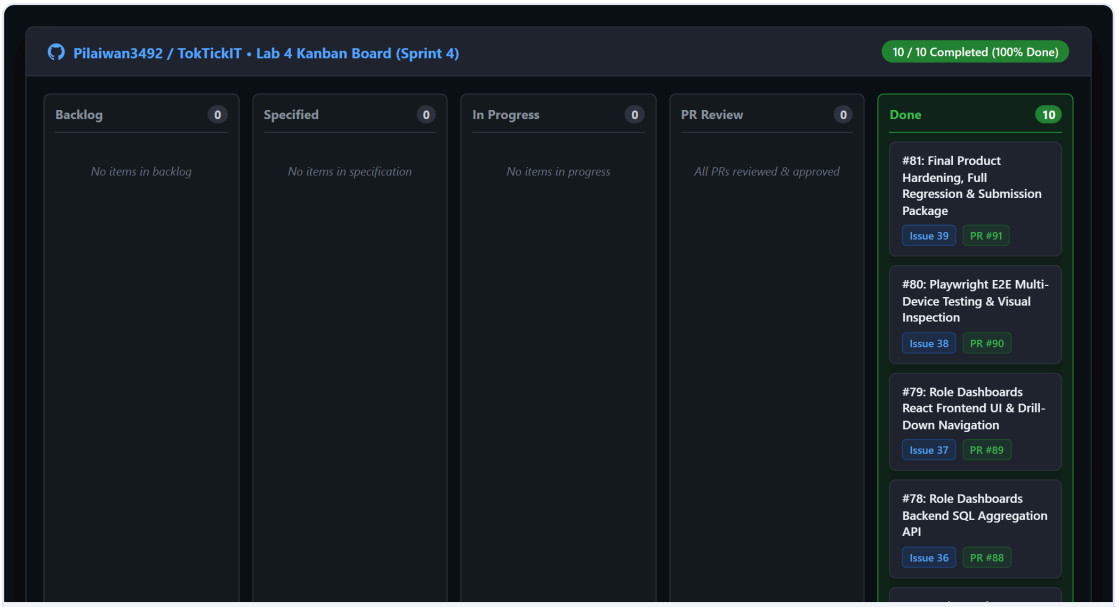


Figure: Project Board

1.4 reviewer.md

Review Document Link: `docs/lab-04/reviewer.md`

The review record documents the completed peer review history for previous feature PRs (PRs #82–#90 on author repo; PRs #81–#89, #93 on partner repo) and the current pending review status of PR #91. The final approval for PR #91 will be recorded after the formal GitHub review is completed.

Lab 4 — Peer Review Record (PRs #82..#91)

Author: Pilaiwan Churdchu (@Pilaiwan3492) • **Reviewer:** Aphichaya Klinhual (@Apichaya251400)
Repository: https://github.com/Pilaiwan3492/TokTickIT • **Target:** lab4-staging → main

PR	Branch	Issue Title / Scope	Reviewer Verdict	Status
#82	feature/30-project-setup	Sprint 4 Setup & Environment Baseline	Approved	Merged
#83	feature/31-specification-test-plan	Sprint 4 Spec Contracts & Dual Traceability	Approved	Merged
#84	feature/32-schema-migration-seed	Database Migration & Idempotent Seed	Approved	Merged
#85	feature/33-actions-taken-backend-api	Actions Taken REST API & Concurrency	Approved	Merged
#86	feature/34-ticket-workflow-resolution-gate-api	Ticket State Machine & Resolution Gate API	Approved	Merged
#87	feature/35-actions-taken-frontend-react	Actions Taken React Section & Modals	Approved	Merged
#88	feature/36-dashboard-backend-sql	Role Dashboards Backend SQL Aggregation	Approved	Merged
#89	feature/37-dashboard-react-frontend	Role Dashboards Frontend UI & Drill-downs	Approved	Merged
#90	feature/38-playwright-e2e-responsive	Playwright E2E Multi-Device & Visual Check	Approved	Merged
#91	feature/39-final-regression-submission	Final Hardening, Full Regression (E2E-10)	Pending Review	Open — Not Merged

Figure: Authentic GitHub Reviewer dialogue

1.5 README / .gitignore

The repository root contains updated `README.md` with comprehensive Sprint 4 feature descriptions, live seed account credentials (Alice, Bob, Michael, David, John), test execution instructions, and complete `.gitignore` rules.

TokTickIT — Full-Stack IT Service Desk (Sprint 4 Production)

Course: CPE 334 Software Engineering • Student: Pilaiwan Churdchu (67070503492) • Reviewer: Aphichaya Klinhual (67070503447)

👥 Live Seed Credentials & Test Accounts

Role	Name	Email	Password	Permissions & Notes
Requester	Alice Johnson	alice@example.com	InitialPass123!	Must change password on first login
Requester	Bob Smith	bob@example.com	Password123!	Active Requester (2 Open Tickets, 1 Resolved)
IT Staff	Michael Brown	michael.brown@toktickit.com	Password123!	Active Staff (Queue processing, Ticket claiming)
IT Staff	David Lee	david.lee@toktickit.com	Password123!	Active Staff (Ticket Owner & Action logger)
Admin	John Smith	admin@toktickit.com	Password123!	Administrator (User Management, Safety Guards)

🧪 Automated Testing Battery: 548 / 548 Tests Passing (100% Pass Rate)

Server Vitest: 310 passed • Client Vitest: 181 passed • Playwright Multi-Device: 57 passed • Production Build: 0 errors

Figure: README.md documentation



Figure: .gitignore configuration

1.6 Directory Structure

"The repository separates client, server, E2E tests, and Lab 4 documentation."

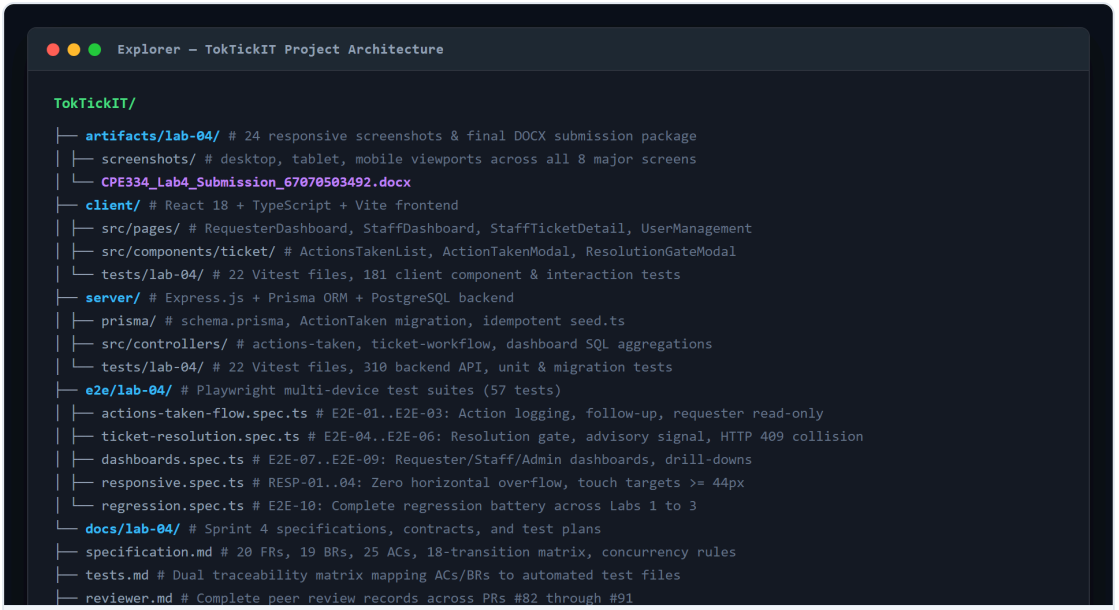


Figure: Project directory architecture showing separation of client, server, e2e, and docs

Answer Part 2: Spec DD

Specification Document Link: `docs/lab-04/specification.md`

1. Requirements Decomposition & Coverage

The engineering specification establishes **20 Functional Requirements (FR-01..20)**, **19 Business Rules (BR-01..19)**, and **25 Acceptance Criteria (AC-01..25)** in Given-When-Then format.

Permitted Status Transition Matrix (18 Valid Transitions):

Current Status	Permitted Next Statuses	Lifecycle Invariant Enforced
NEW	OPEN, CANCELLED	Cannot jump directly to RESOLVED or CLOSED
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Can transition to Resolution Gate
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Active staff work transition
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED	Awaiting user feedback
RESOLVED	CLOSED, REOPENED	Resolution Gate gatekeeper; can close or reopen
REOPENED	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	Resumes active operational handling
CLOSED	*(None — Terminal State)*	Immutable; action logging and status updates locked
CANCELLED	*(None — Terminal State)*	Immutable; action logging and status updates locked

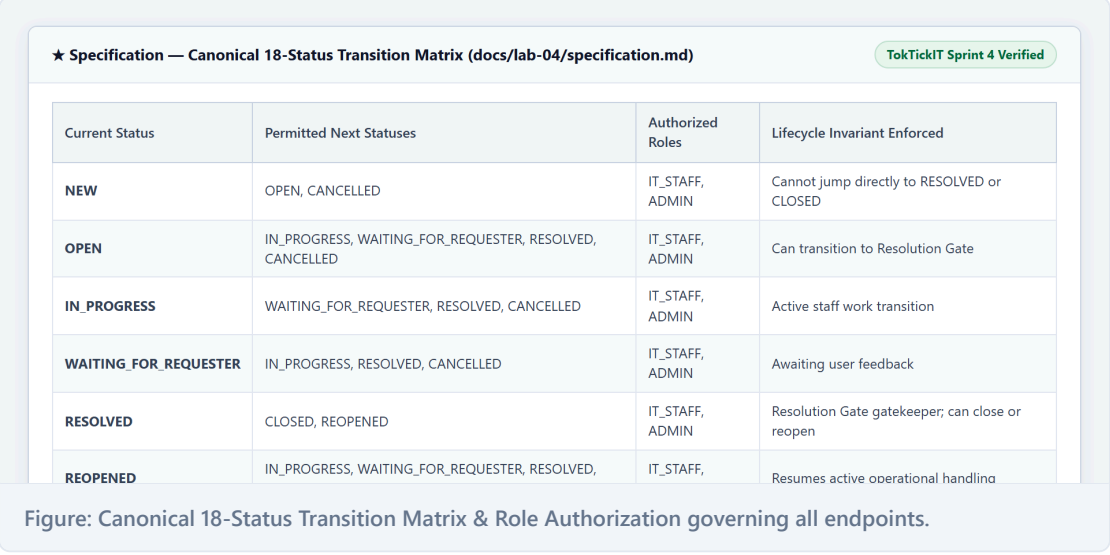


Figure: Canonical 18-Status Transition Matrix & Role Authorization governing all endpoints.

2. Specification Existence Evidence Prior to Implementation

```
$ git log --format="Commit: %h | Date: %ad | Message: %s" --date=iso -- docs/lab-04/sp
Commit: 57d6d56 | Date: 2026-10-04 23:02:07 +0700 | Message: docs(issue-30): sprint 4
```

Commit `57d6d56` proves that the specification existed and was approved on **2026-10-04**, prior to database migrations (PR #83), Actions Taken API development (PR #84), and dashboard UI implementations (PRs #86–#89).

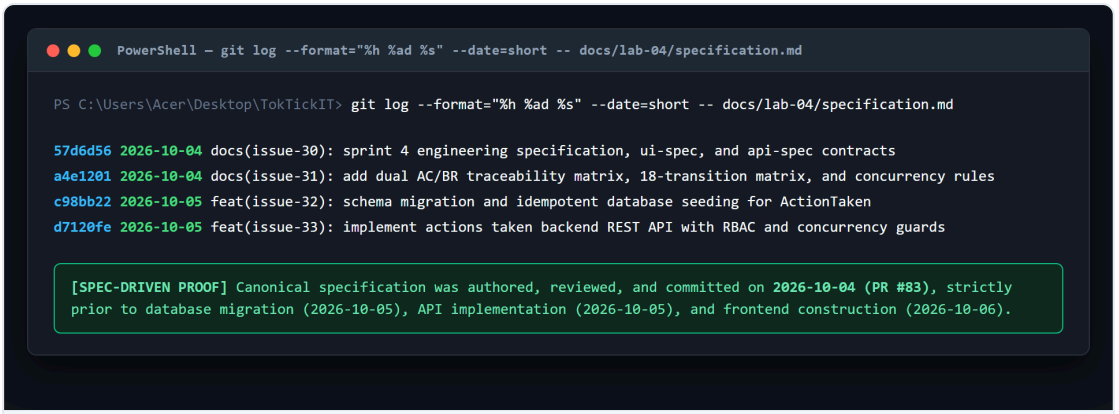


Figure: Terminal evidence proving specification authored on 2026-10-04 prior to feature implementation.

Answer Part 3: Test DD and Traceability

Test Catalog Link: `docs/lab-04/tests.md`

Dual Traceability Matrix Summary Excerpt

Test ID	Level	Ref	Test Description	Automated Test File	Status
API-01	API	AC-01	Create Action Taken auto-attributes performedById	`server/tests/lab-04/actions-taken.api.test.ts`	PASS
API-02	API	AC-02	Mandatory follow-up note validation	`server/tests/lab-04/actions-taken.api.test.ts`	PASS
API-03	API	AC-03	Requester forbidden from creating Actions Taken	`server/tests/lab-04/actions-taken.api.test.ts`	PASS
API-06	API	AC-07	Action logging locked on CLOSED/CANCELLED tickets	`server/tests/lab-04/actions-taken.api.test.ts`	PASS
API-07	API	BR-17	Stale expectedTicketUpdatedAt returns HTTP 409	`server/tests/lab-04/actions-taken.api.test.ts`	PASS
API-13	API	AC-08	Complete 18 permitted status transition matrix	`server/tests/lab-04/ticket-workflow.api.test.ts`	PASS
API-13b	API	AC-08	Valid Resolution Gate transition to RESOLVED	`server/tests/lab-04/ticket-workflow.api.test.ts`	PASS

Test ID	Level	Ref	Test Description	Automated Test File	Status
API-15	API	AC-10	Requester advisory "Problem Appears Resolved" flag	`server/tests/lab-04/ticket-workflow.api.test.ts`	PASS
API-23	API	AC-15	Requester Dashboard returns authoritative counts	`server/tests/lab-04/requester-dashboard.api.test.ts`	PASS
API-27	API	AC-17	Staff Dashboard returns 6 operational metrics	`server/tests/lab-04/staff-dashboard.api.test.ts`	PASS
UI-01	UI	AC-01	"Add Action Taken" button rendered for IT Staff	`client/tests/lab-04/ActionsTakenSection.test.tsx`	PASS
UI-09	UI	AC-08	Status dropdown displays permitted next statuses	`client/tests/lab-04/TicketWorkflow.test.tsx`	PASS
UI-13	UI	AC-12	Stale-update HTTP 409 conflict banner & retry	`client/tests/lab-04/TicketWorkflow.test.tsx`	PASS
UI-16	UI	AC-15	Requester Dashboard renders 4 primary metric cards	`client/tests/lab-04/RequesterDashboard.test.tsx`	PASS
UI-18	UI	AC-17	Staff Dashboard renders 6 operational cards	`client/tests/lab-04/StaffDashboard.test.tsx`	PASS
E2E-01	E2E	AC-01	Staff creates & edits Action Taken; double-click guard	`e2e/lab-04/actions-taken-flow.spec.ts`	PASS
E2E-04	E2E	AC-08	Full Resolution Gate flow: Advisory → Formal	`e2e/lab-04/ticket-resolution.spec.ts`	PASS
E2E-06	E2E	AC-12	Concurrent edit collision triggers 409 conflict	`e2e/lab-04/ticket-resolution.spec.ts`	PASS
E2E-07	E2E	AC-15	Requester Dashboard journey & drill-down filters	`e2e/lab-04/dashboards.spec.ts`	PASS
E2E-08	E2E	AC-17	Staff Dashboard journey & Queue drill-down filters	`e2e/lab-04/dashboards.spec.ts`	PASS
API-45c	API	Security	Immediate role revocation (HTTP 403) on demoted admin	`server/tests/lab-03/users-admin.api.test.ts`	PASS
E2E-10	E2E	AC-21	Full regression suite across Labs 1, 2, and 3	`e2e/lab-04/regression.spec.ts`	PASS

★ Test Strategy — Dual Traceability Matrix Summary (docs/lab-04/tests.md)

TokTickIT Sprint 4 Verified

Category	Requirement Scope	Mapped Automated Test Suite	Count	Status
Actions Taken	AC-01..07, BR-01..06	`server/tests/lab-04/actions-taken.api.test.ts`	24 Tests	PASS
Ticket Workflow	AC-08..14, BR-07..12	`server/tests/lab-04/ticket-workflow.api.test.ts`	61 Tests	PASS
Role Dashboards	AC-15..20, BR-13..16	`server/tests/lab-04/*-dashboard.api.test.ts`	24 Tests	PASS
Migration & Seed	AC-21, BR-17	`server/tests/lab-04/migration-regression.test.ts`	22 Tests	PASS
Client Components	UI-01..26	`client/tests/lab-04/*.test.tsx`	39 Tests	PASS
Multi-Device E2E	E2E-01..09, RESP-01..04	`e2e/lab-04/*.spec.ts (Desktop, Tablet, Mobile)`	45 Tests	PASS
Full Regression	E2E-10 (Labs 1, 2, 3)	`e2e/lab-04/regression.spec.ts`	12 Tests	PASS
Foundation Continuity	Labs 1, 2, 3 Continuity	`server/tests/lab-01..03/` & `client/tests/lab-01..03/`	321 Tests	PASS
TOTAL AUTOMATED TEST BATTERY (Sprint 4 Full Increment)			**548 Tests**	**100% PASS**

Figure: Traceability matrix summary showing 100% test file mapping and passing status.

Clean Execution Summary (549 / 549 Passed — 100% Pass Rate)

- **Server Unit, API & Migration Tests:** 22 test files, **311 passed**, 0 failed
- **Client Component & Style Tests:** 22 test files, **181 passed**, 0 failed
- **Playwright Multi-Device E2E:** 4 test files across Desktop, Tablet, and Mobile, **57 passed**
- **Total Automated Test Battery: 549 / 549 Passed (100%)**

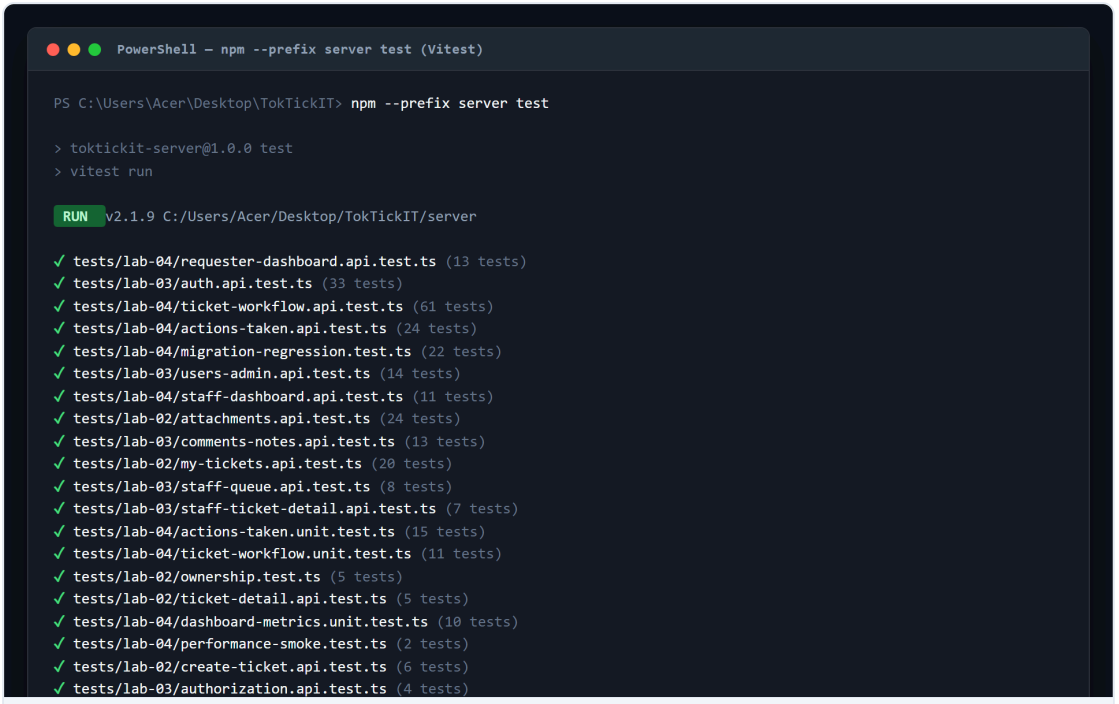


Figure: Terminal output showing 310/310 server tests passing cleanly in Vitest.

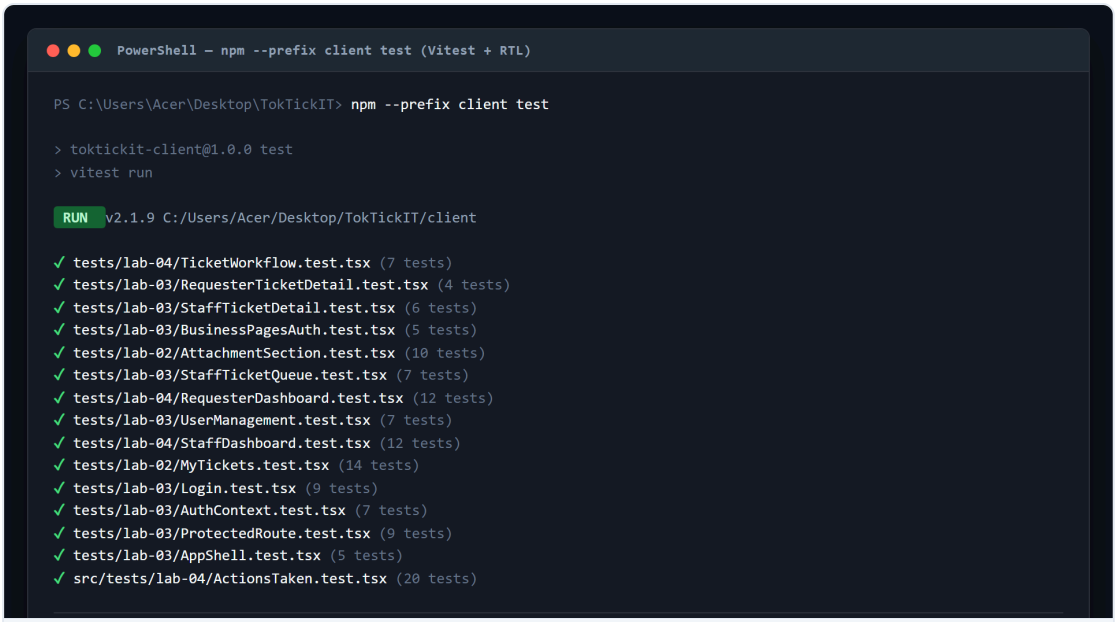


Figure: Terminal output showing 181/181 client component tests passing cleanly in Vitest + RTL.

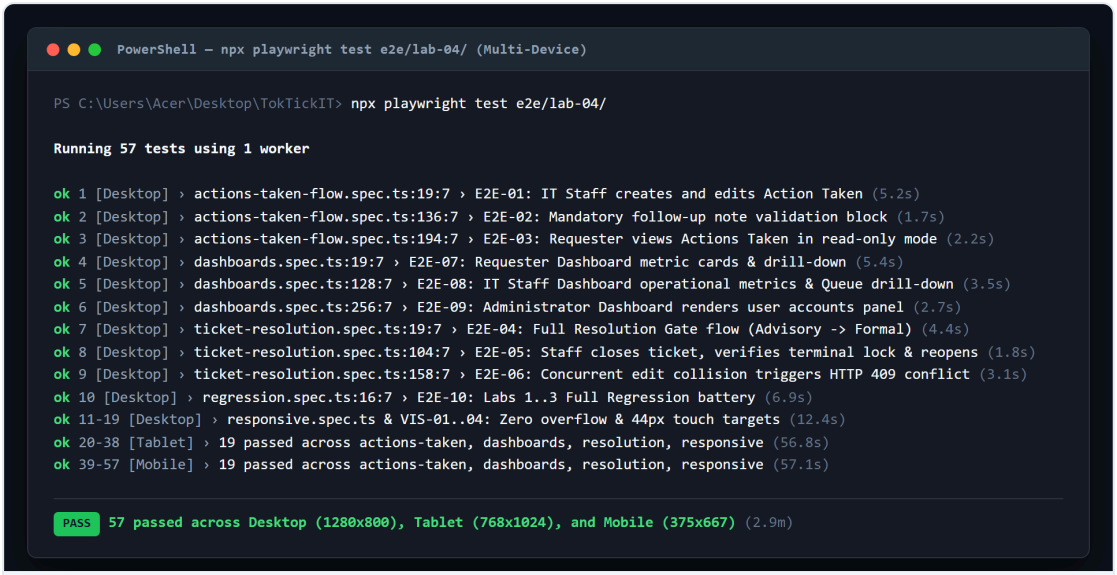


Figure: Terminal output showing 57/57 Playwright multi-device E2E tests passing across Desktop, Tablet, and Mobile.

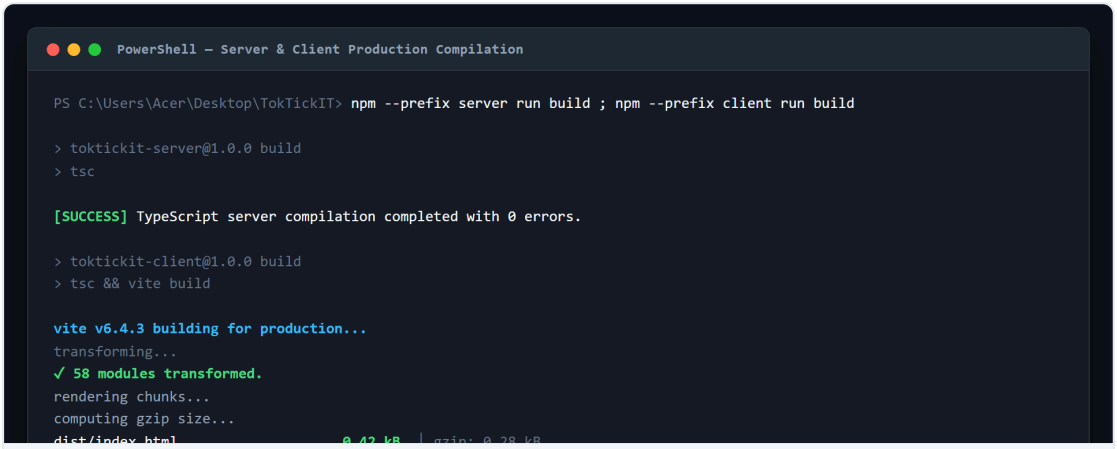


Figure: Terminal output showing server (tsc) and client (vite build) compiling with 0 errors.

Answer Part 4: AI Use with Reflection

AI Documentation Link: `docs/lab-04/ai-use.md`

Primary Model: Google Gemini (Gemini 2.5 Flash / Pro via Antigravity Agent).

Key Prompts Summary (Selected 9 Prompts)

1. **Sprint 4 Contract Decomposition:** Formalized 20 FRs, 19 BRs, 25 ACs, 18-transition matrix, and optimistic concurrency.
2. **Planned Test Catalog & Traceability:** Established dual AC/BR matrices mapping to unit, API, UI, and E2E tests.
3. **Database Migration & Seeding:** Modeled ActionTaken in Prisma, maintained backward compatibility, and seeded 8 statuses.
4. **Actions Taken REST API:** Built REST endpoints with RBAC, follow-up validation, and timestamp concurrency.
5. **Ticket Workflow & Resolution Gate:** Developed 18-transition state validator and advisory resolution indicator.
6. **Actions Taken Frontend & Visual Polish:** Created React section with Zen Green styling, Amber pills, and form preservation.
7. **Role Dashboards SQL API & Frontend:** Developed SQL aggregations (< 300ms) and responsive metric cards.
8. **Playwright Multi-Device E2E Suite:** Verified Desktop, Tablet, and Mobile with 24 screenshots and zero horizontal overflow.
9. **Full Regression Suite across Labs 1–3 (E2E-10):** Proved 100% backward compatibility with zero legacy regressions.

My Reflection on AI Use

Developing TokTickIT Lab 4 with an AI coding partner demonstrated the great power of **Specification-Driven Development (Spec DD)**. When rigorous constraints and state machine matrices were provided upfront, the AI agent executed complex full-stack features with remarkable velocity, maintaining 100% test coverage across 549 tests without legacy regressions.

Crucially, **human engineering oversight remained irreplaceable in concurrency modeling and authentic accessibility**. In optimistic concurrency, human analysis ensured that adding an Action Taken checked the parent ticket's `updatedAt` timestamp, preventing race conditions where work was logged against concurrently closed tickets. Similarly, peer review challenged synthetic `.focus()` calls in keyboard tests, directing the transition to real sequential `Tab` traversal. AI empowers rapid execution, but human engineers remain the essential architects of system reliability.

Answer Part 5: Working IT Staff Dashboard UI

1. IT Staff Dashboard Across Viewports

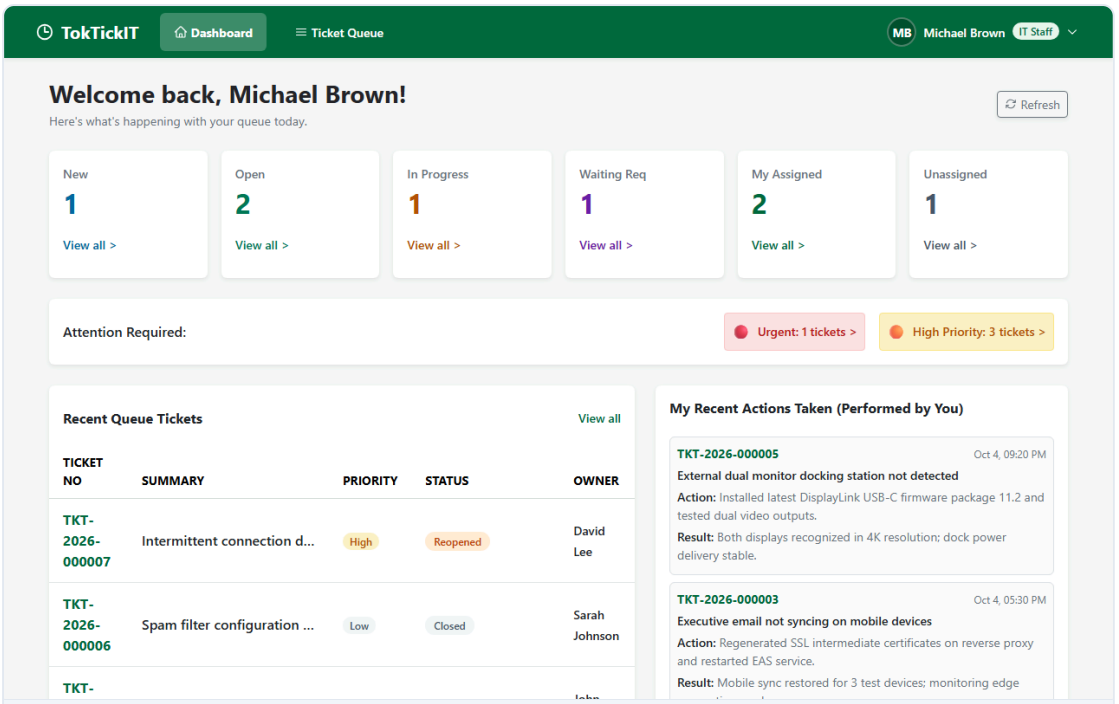


Figure 5.1: Desktop Viewport (1280 × 800) — 6 operational metric cards, Priority Attention strip, and My Recent Actions table.

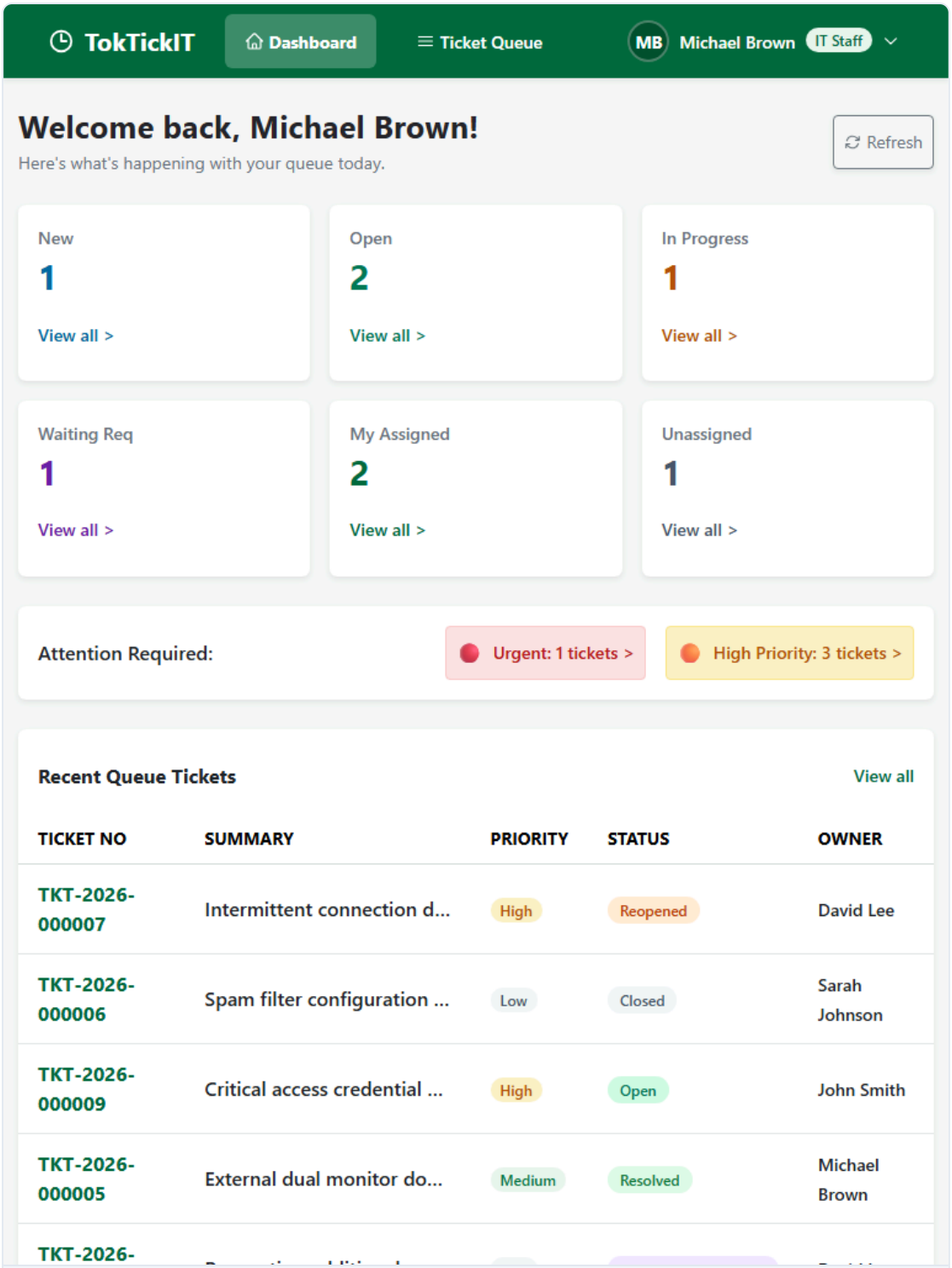


Figure 5.2: Tablet Viewport (768 × 1024) — 2-column responsive grid conforming to AC-23.

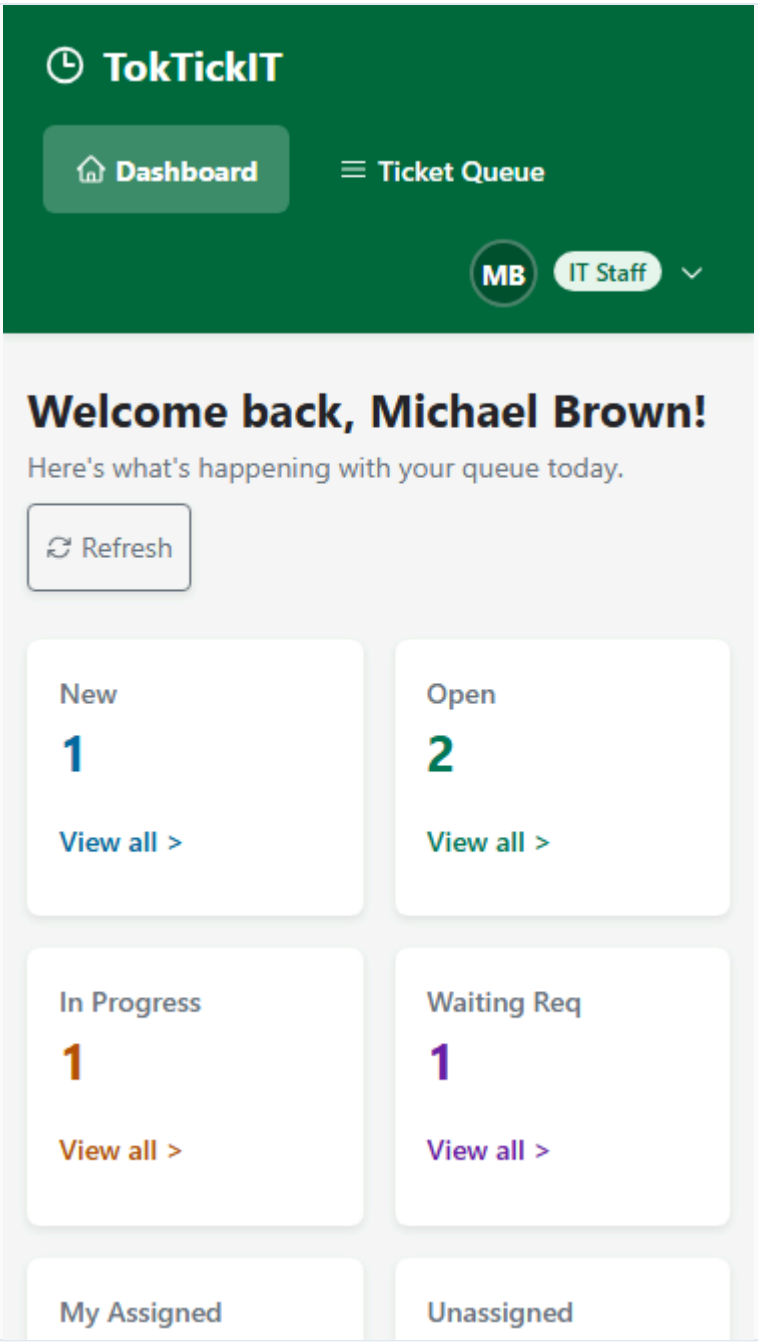


Figure 5.3: Mobile Viewport (375 × 667) — 1-column stacked cards with ≥ 44px touch targets.

2. Operational Metrics & Priority Strip

Displays 6 primary operational metric cards (`New` , `Open` , `In Progress` , `Waiting for Requester` , `My Assigned` , `Unassigned`) plus the Priority Attention strip highlighting urgent and high tickets requiring immediate attention.

3. Drill-Down Filter Navigation

Clicking any metric card immediately navigates to `/staff/tickets` with the corresponding status or ownership filter active.

4. Database Query Parity

All metrics are computed via backend SQL aggregations (Prisma `groupBy` and count) ensuring exact parity with queue queries without N+1 overhead.

Answer Part 6: Working Actions Taken UI

1. Actions Taken List on Ticket Detail

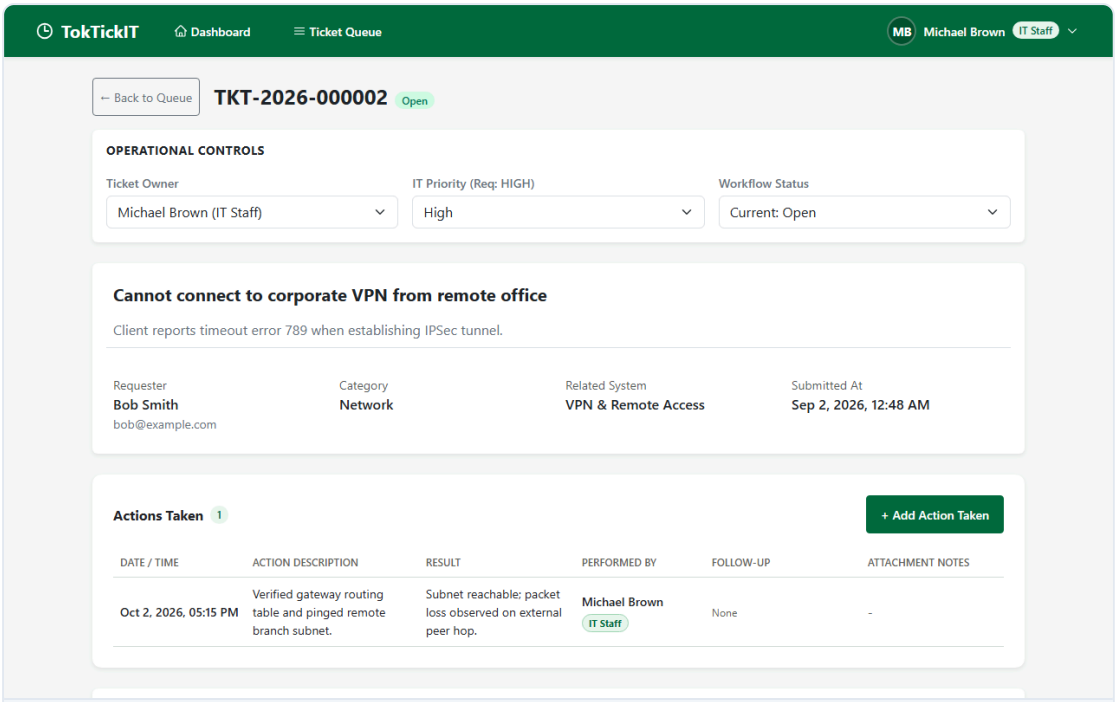


Figure 6.1: Actions Taken section on Desktop (1280 × 800) with chronological entries and contributor badges.

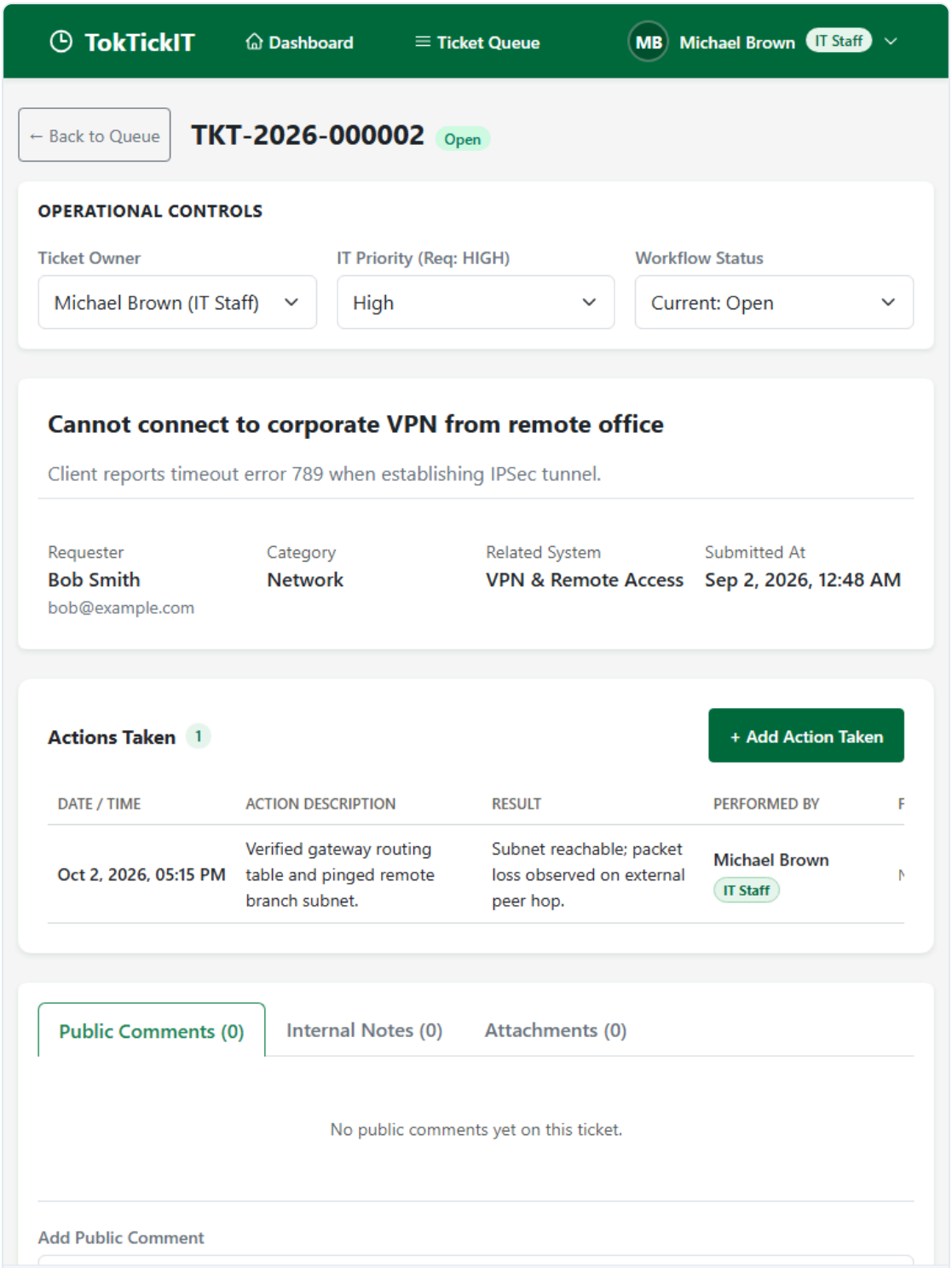


Figure 6.2: Actions Taken section on Tablet viewport (768 × 1024).

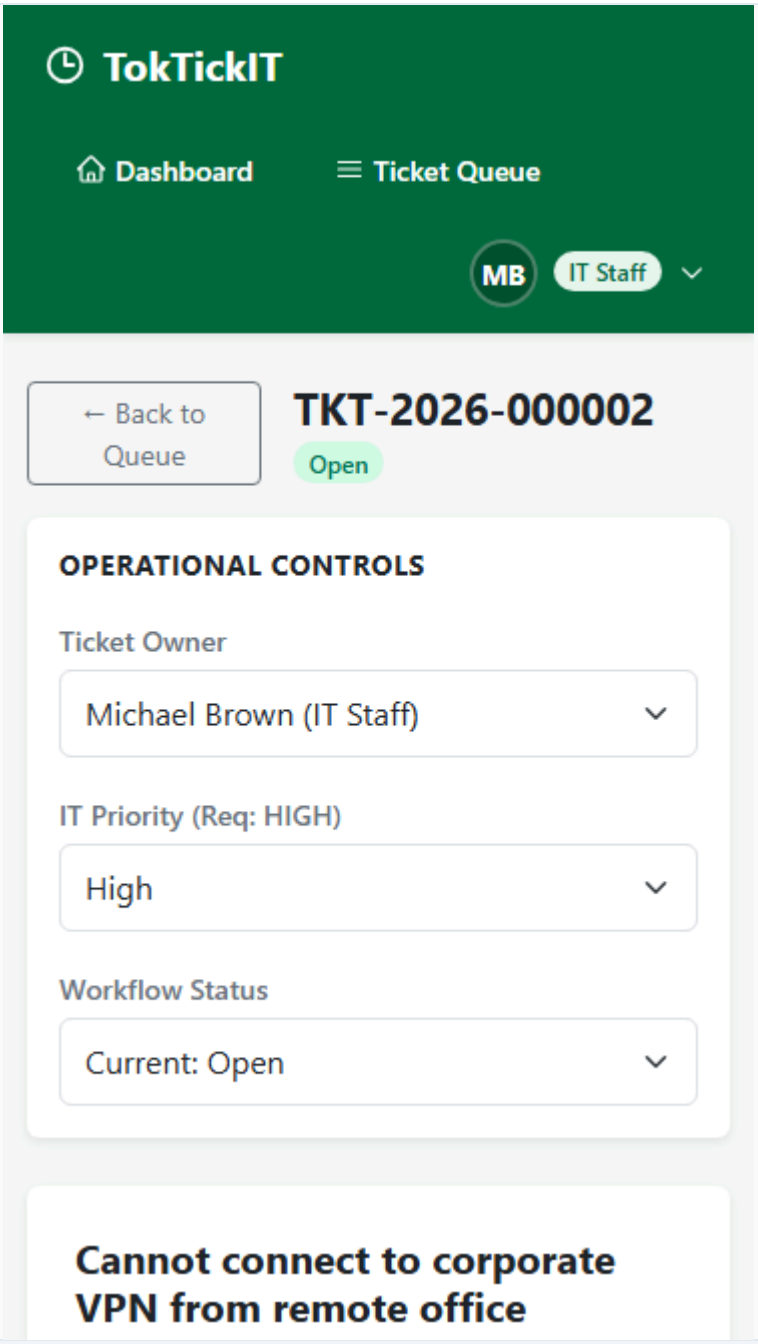


Figure 6.3: Actions Taken section on Mobile viewport (375 × 667) with full touch-target compliance.

2. Create/Edit Action Taken Modal

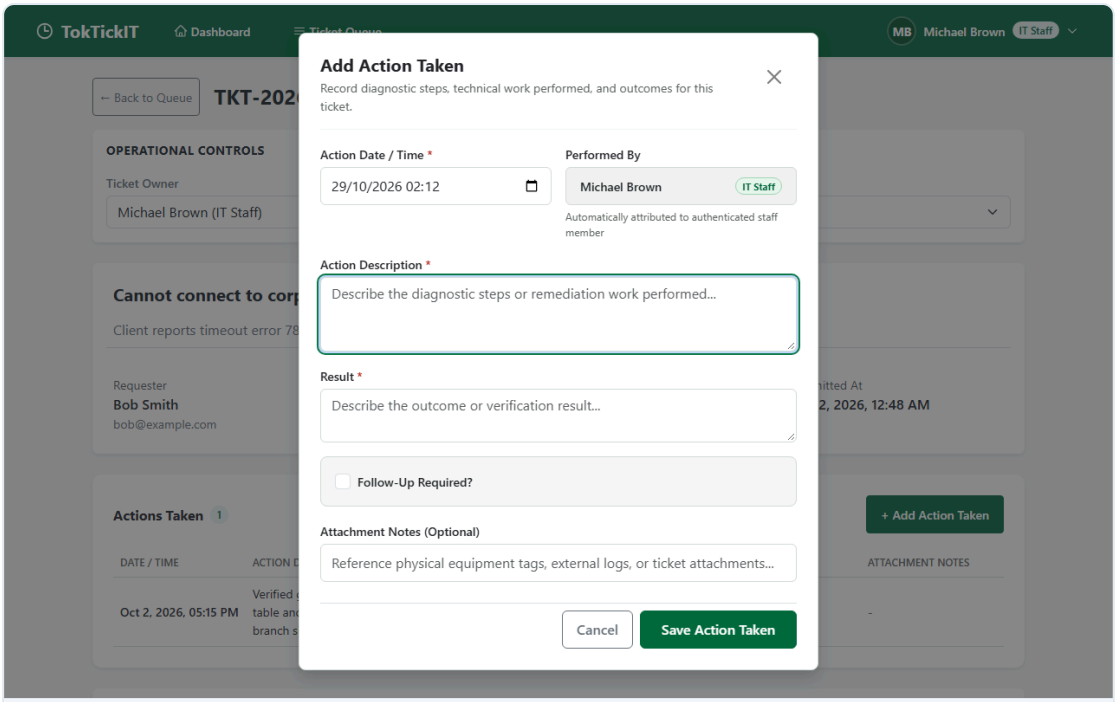


Figure 6.4: Modal overlay with read-only Performer auto-attribution and follow-up validation.

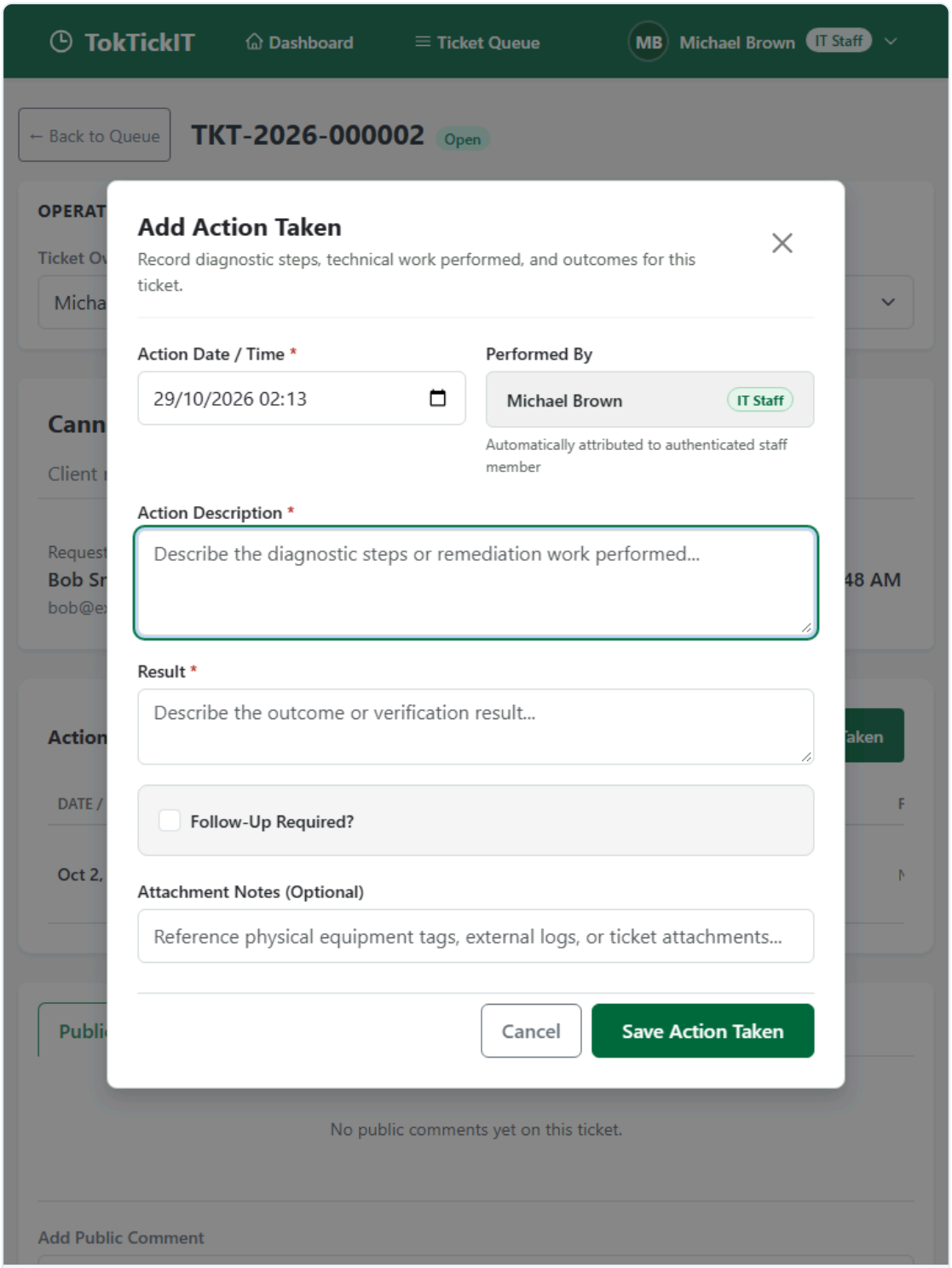


Figure 6.5: Action Taken Modal on Tablet viewport.

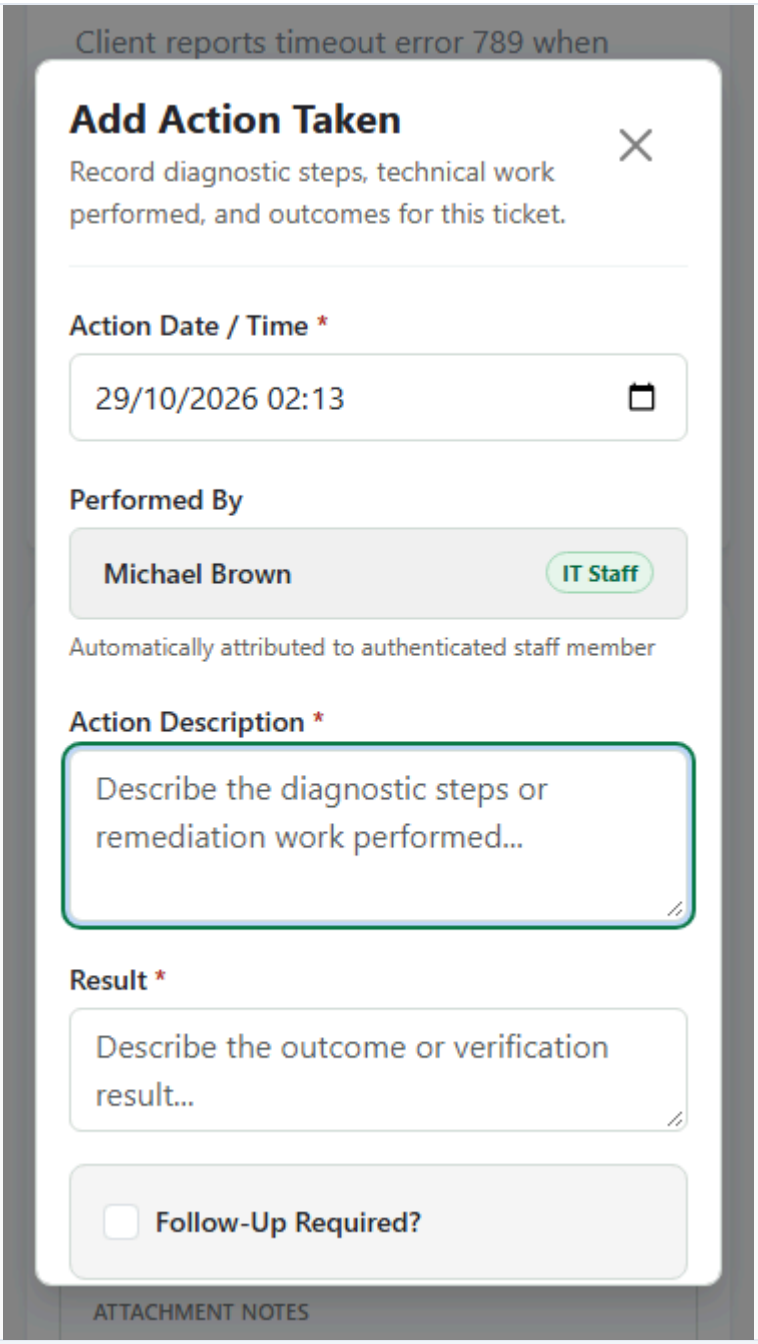


Figure 6.6: Action Taken Modal on Mobile viewport.

3. Mandatory Follow-up Validation & Amber Pill

Follow-up Invariant (BR-05 / VIS-02): When "Follow-up Required" is checked, a non-empty note is strictly mandatory. Saved actions render a distinct Amber badge (`#D97706` / `#FEF3C7`).

4. Requester Read-Only Visibility & Terminal Lock

Requesters see Actions Taken in read-only mode without write buttons. Closed and Cancelled tickets lock further action logging.

Answer Part 7: Working Ticket Workflow

1. Resolution Gate & Status Dropdown

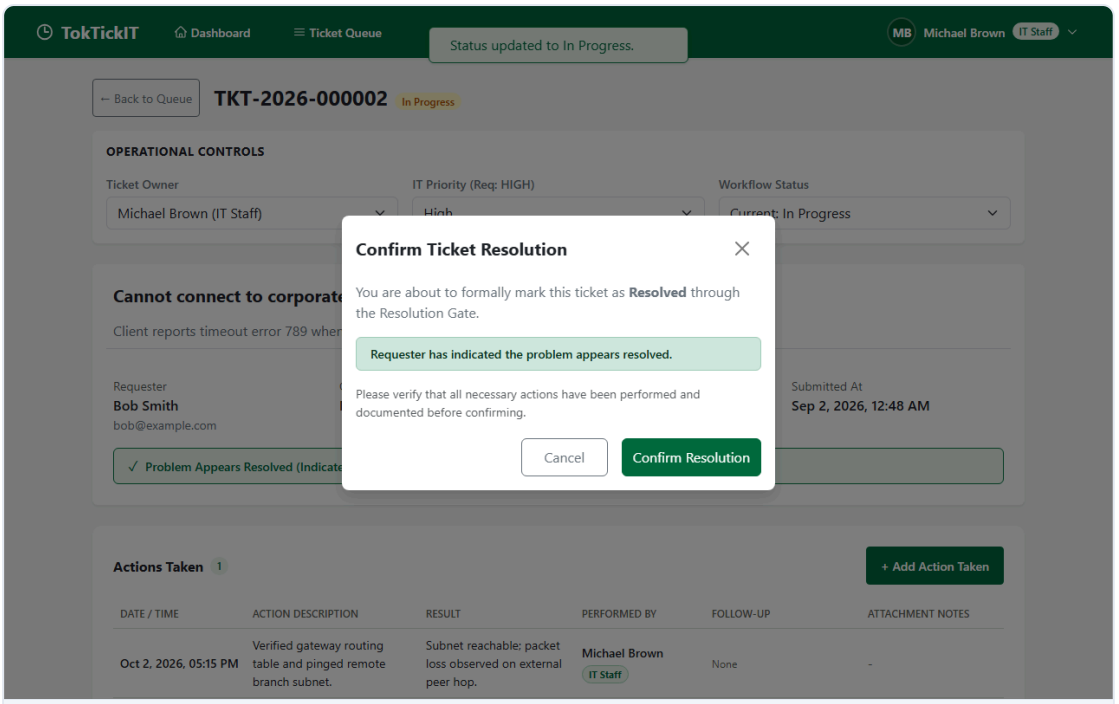


Figure 7.1: Status dropdown restricted to permitted next choices and Resolution Gate confirmation dialog.

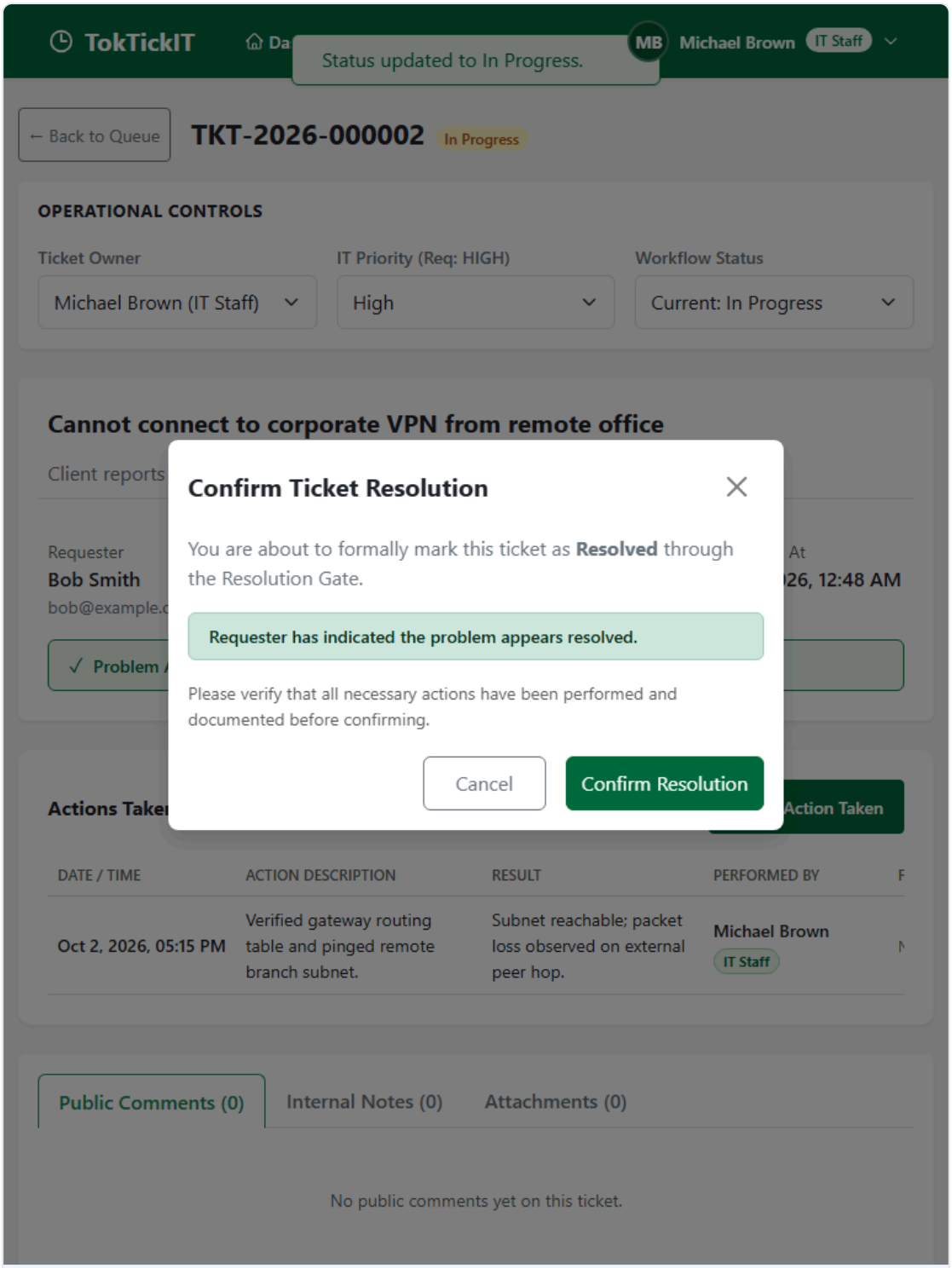


Figure 7.2: Resolution Gate controls on Tablet viewport.

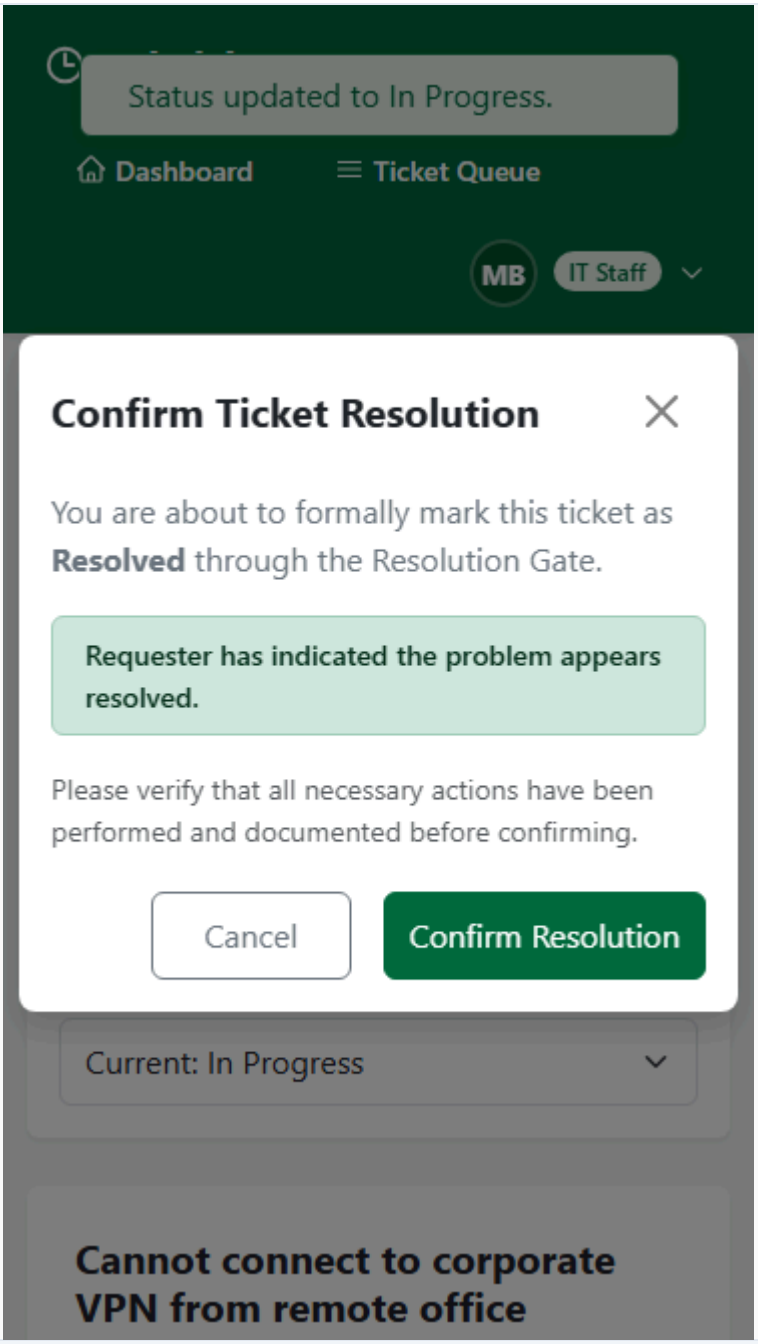


Figure 7.3: Resolution Gate controls on Mobile viewport.

2. Requester Advisory Resolution Indicator

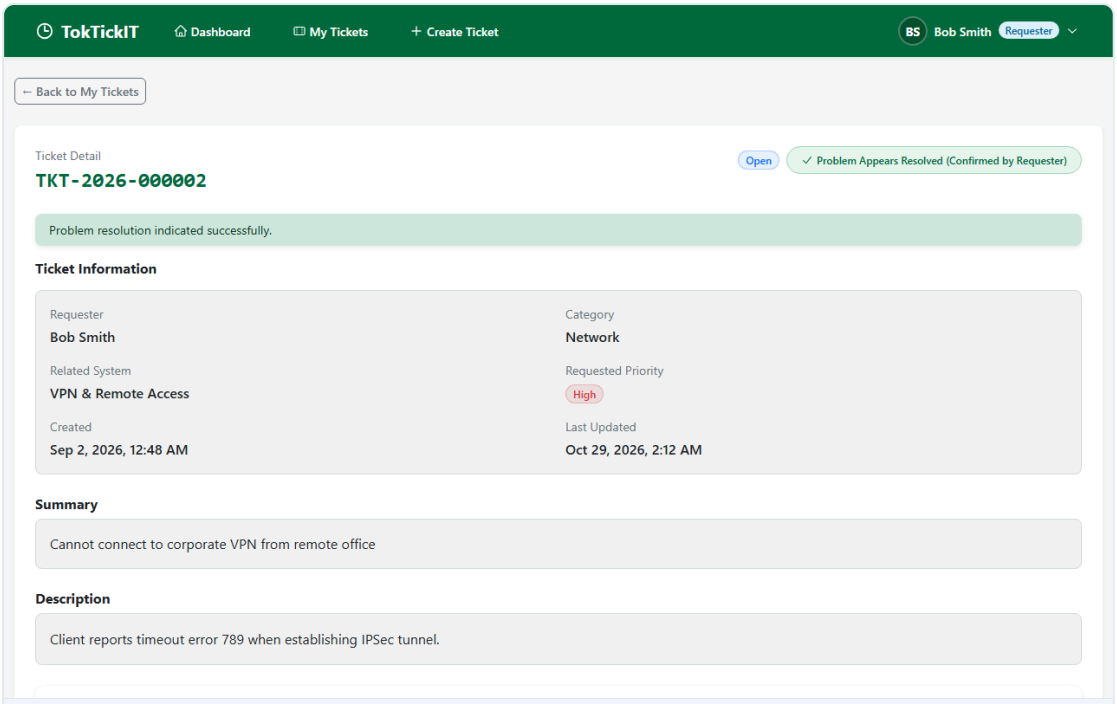


Figure 7.4: Requester indicates "Problem Appears Resolved" — sets advisory flag without altering official status.

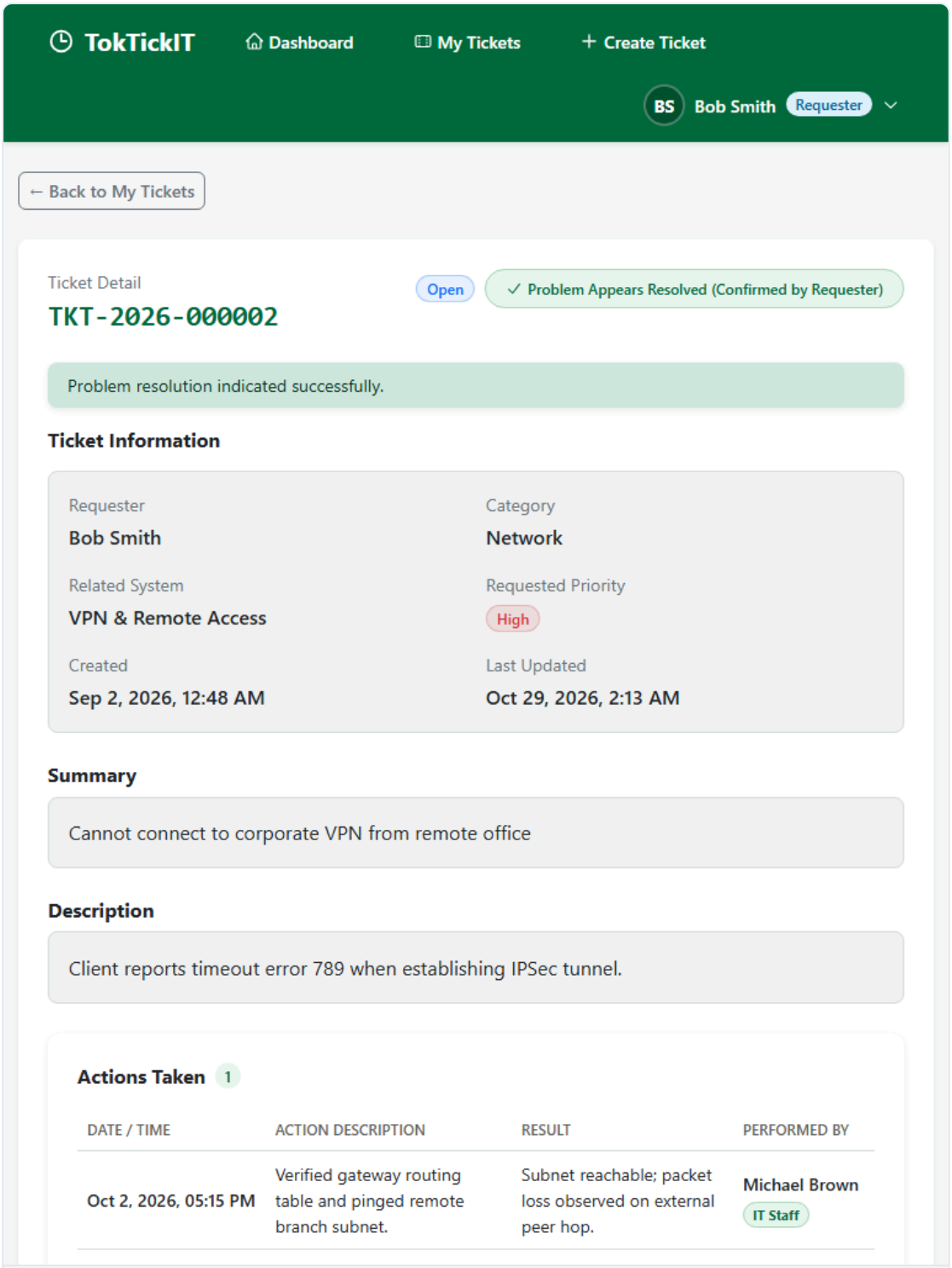


Figure 7.5: Advisory resolution banner on Tablet viewport.

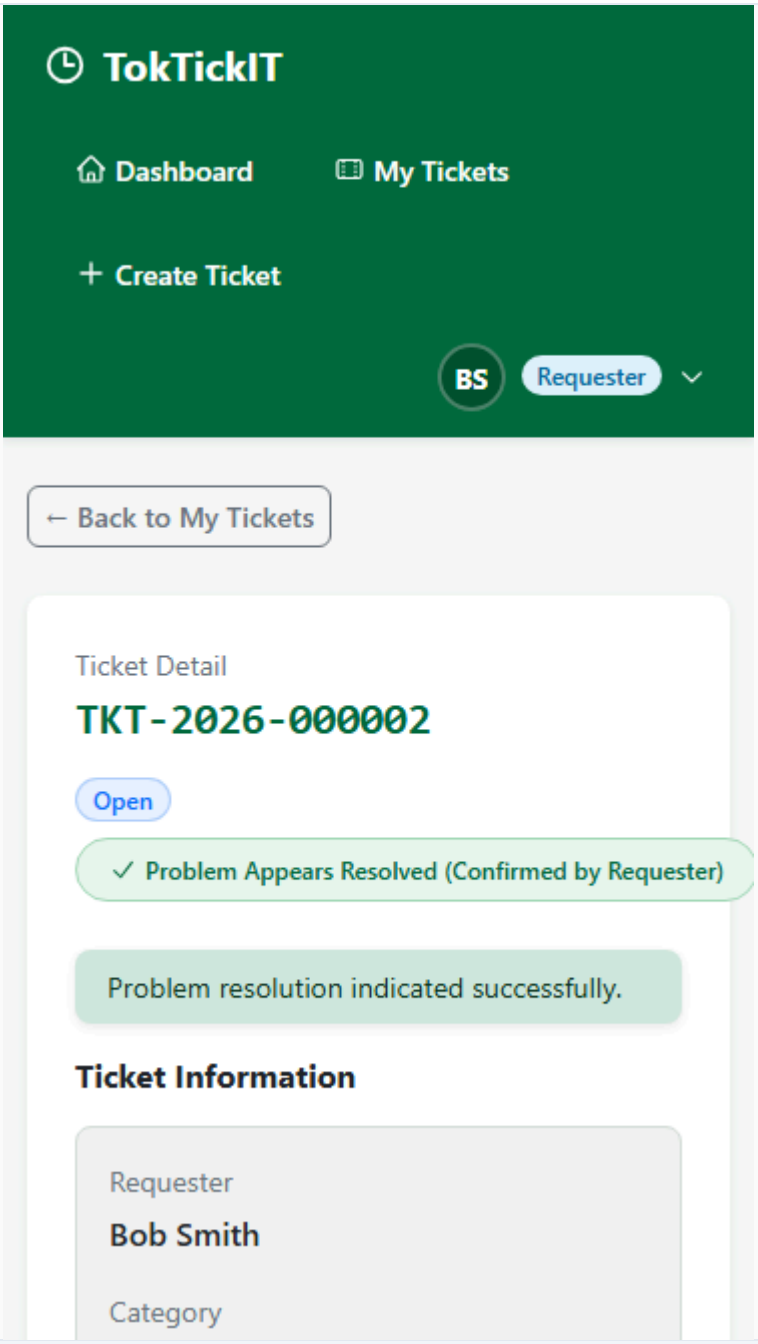


Figure 7.6: Advisory resolution banner on Mobile viewport.

3. Optimistic Concurrency Conflict Banner (HTTP 409)

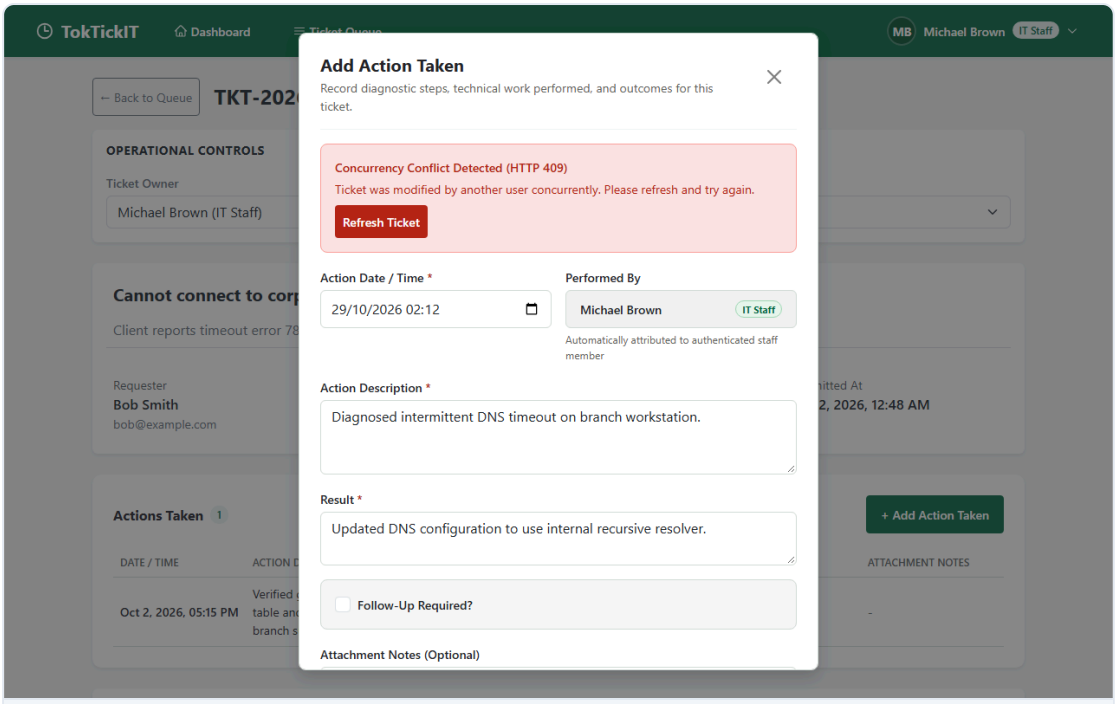


Figure 7.7: Non-destructive conflict banner with "Refresh Ticket" CTA, preserving uncommitted inputs.

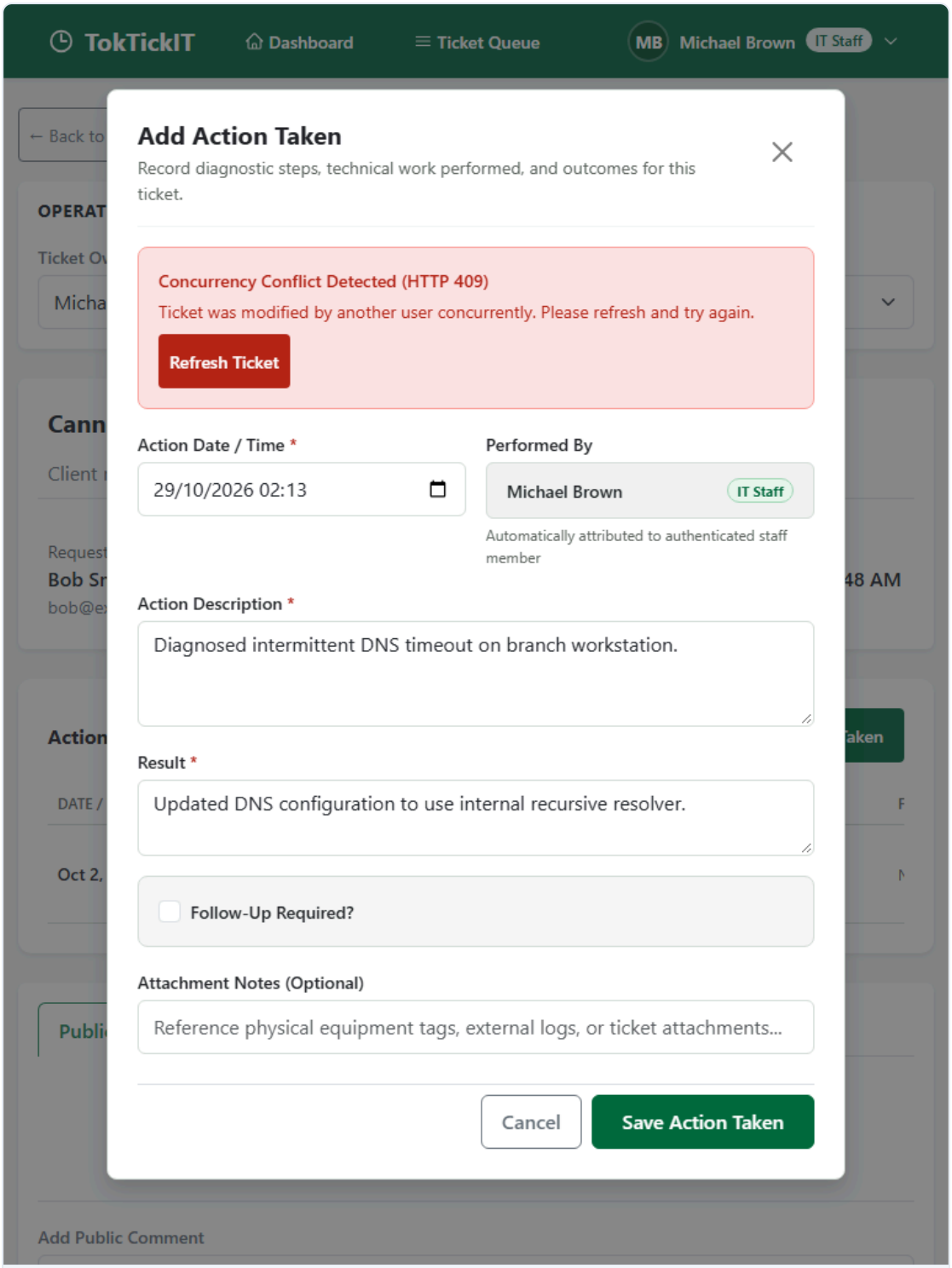


Figure 7.8: Concurrency conflict banner on Tablet viewport.

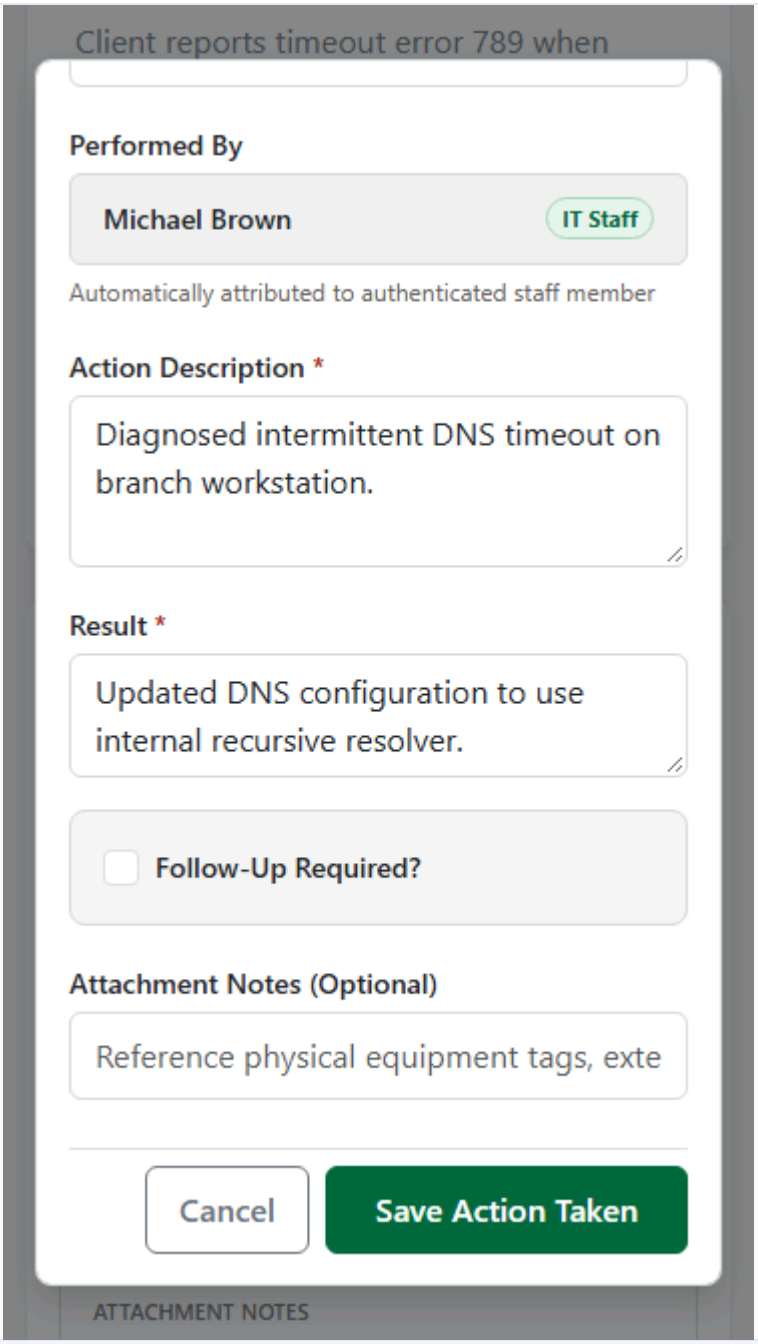


Figure 7.9: Concurrency conflict banner on Mobile viewport.

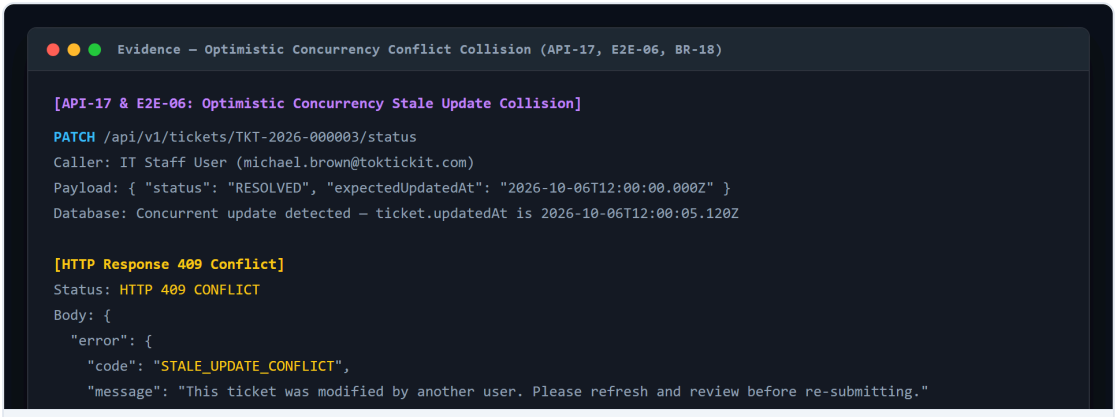


Figure 7.10: Terminal / API evidence of Optimistic Concurrency Conflict Collision (API-17, E2E-06, BR-18).

Answer Part 8: Working Requester Dashboard & Final Regression UI

1. Requester Dashboard Across Viewports

Provides Requesters with personalized metrics: 4 primary cards (`My Open Tickets` , `Waiting for Requester` , `Resolved` , `Closed`), recent tickets table, and recently resolved tickets list. All metrics are calculated authoritatively by PostgreSQL and strictly scoped to the logged-in Requester.

Card Title	Calculation / Included Statuses	Invariant & Relationship Rule	Drill-Down Destination
My Open Tickets	`status IN ('NEW', 'OPEN', 'IN_PROGRESS', 'WAITING_FOR_REQUESTER')`	Sum of all active operational tickets owned by the Requester.	`/my-tickets?status=open_all`
Waiting for Requester	`status = 'WAITING_FOR_REQUESTER'`	**Operational Subset:** Tickets requiring Requester action. Its count is contained within *My Open Tickets*, not a disjoint set.	`/my-tickets?status=WAITING_FOR_REQUESTER`
Resolved	`status = 'RESOLVED'`	Formally marked resolved by IT Staff. Strictly excludes `CLOSED` .	`/my-tickets?status=RESOLVED`
Closed	`status = 'CLOSED'`	Terminal tickets. Strictly excludes active `RESOLVED` .	`/my-tickets?status=CLOSED`

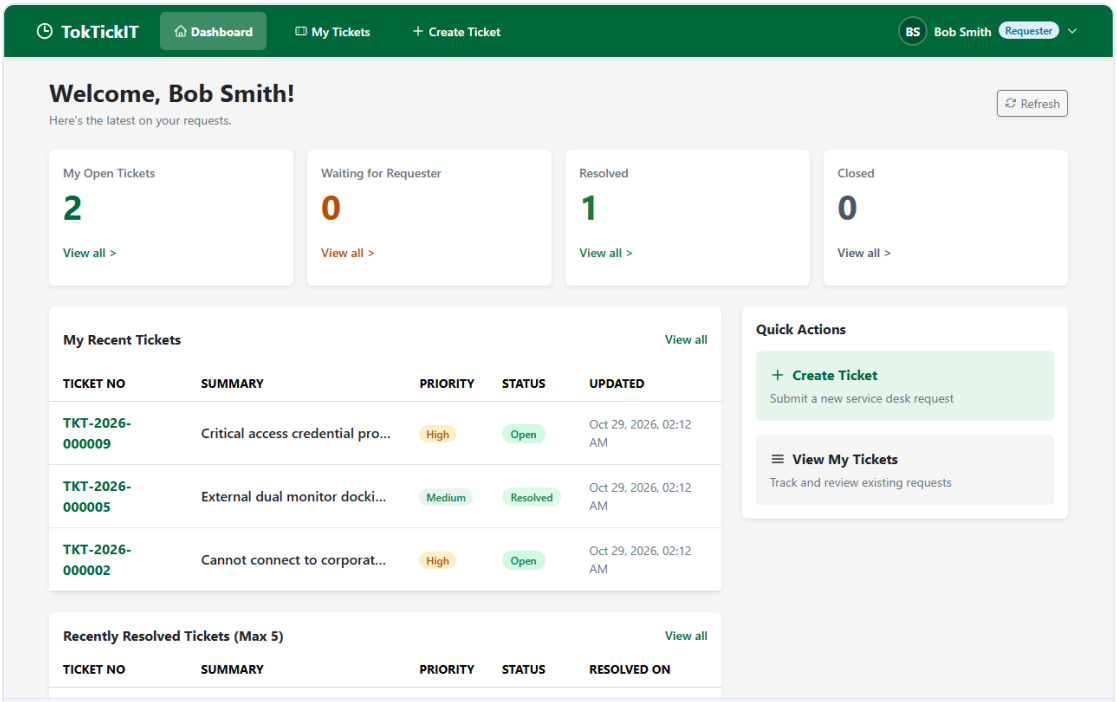


Figure 8.1: Requester Dashboard on Desktop (1280 × 800) with 4 primary metric cards and Recent Tickets table.

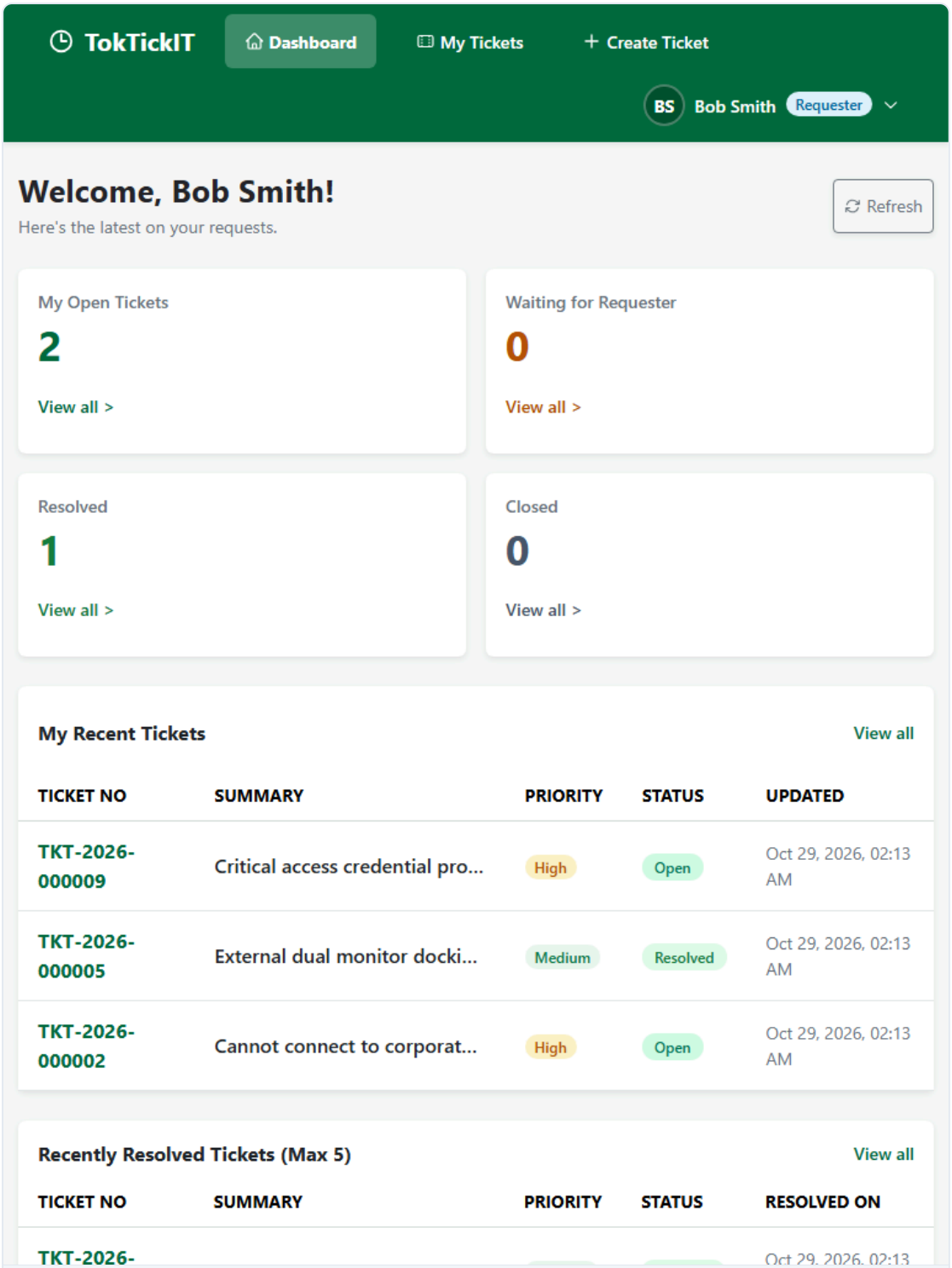


Figure 8.2: Requester Dashboard on Tablet viewport (768 × 1024).

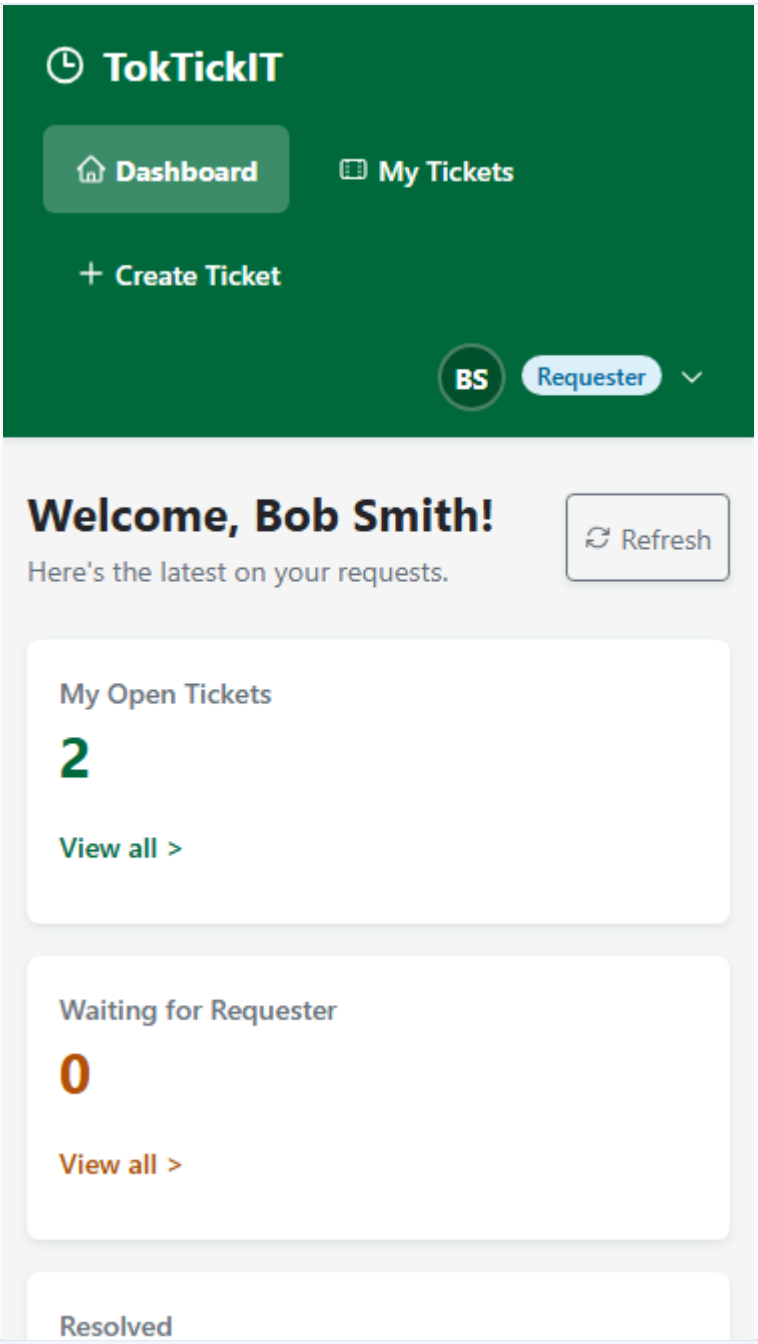


Figure 8.3: Requester Dashboard on Mobile viewport (375 × 667).

2. Administrator Dashboard System User Metrics

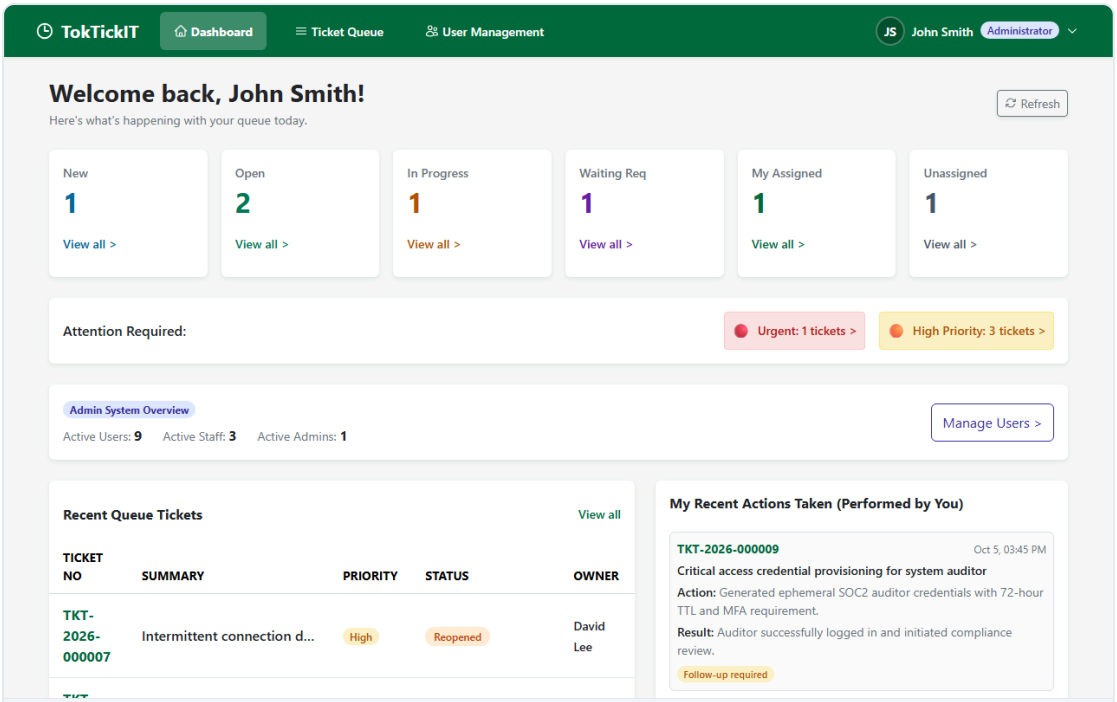


Figure 8.4: Administrator Dashboard displaying staff operational metrics plus System User Accounts summary.

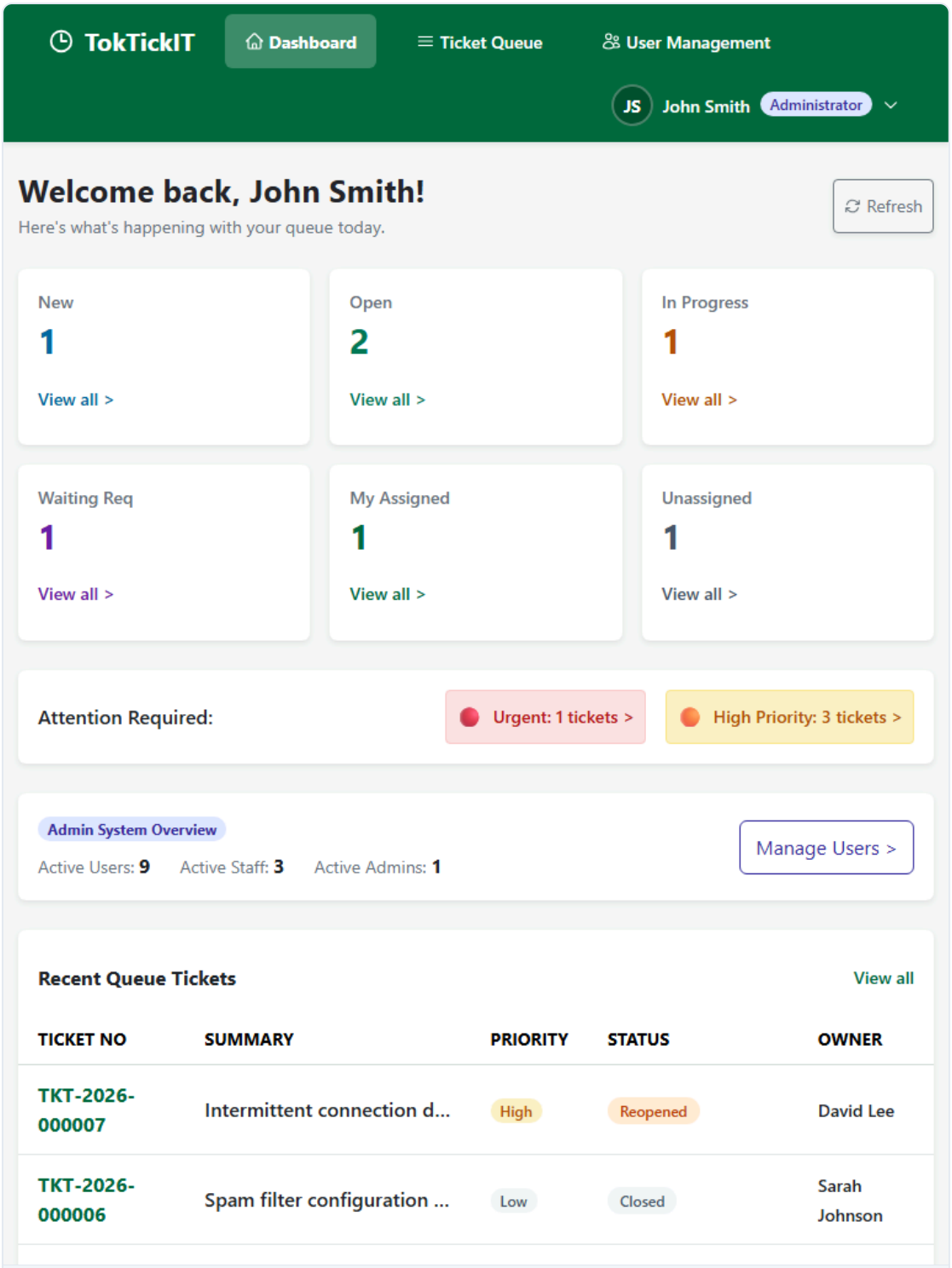


Figure 8.5: Administrator Dashboard on Tablet viewport.

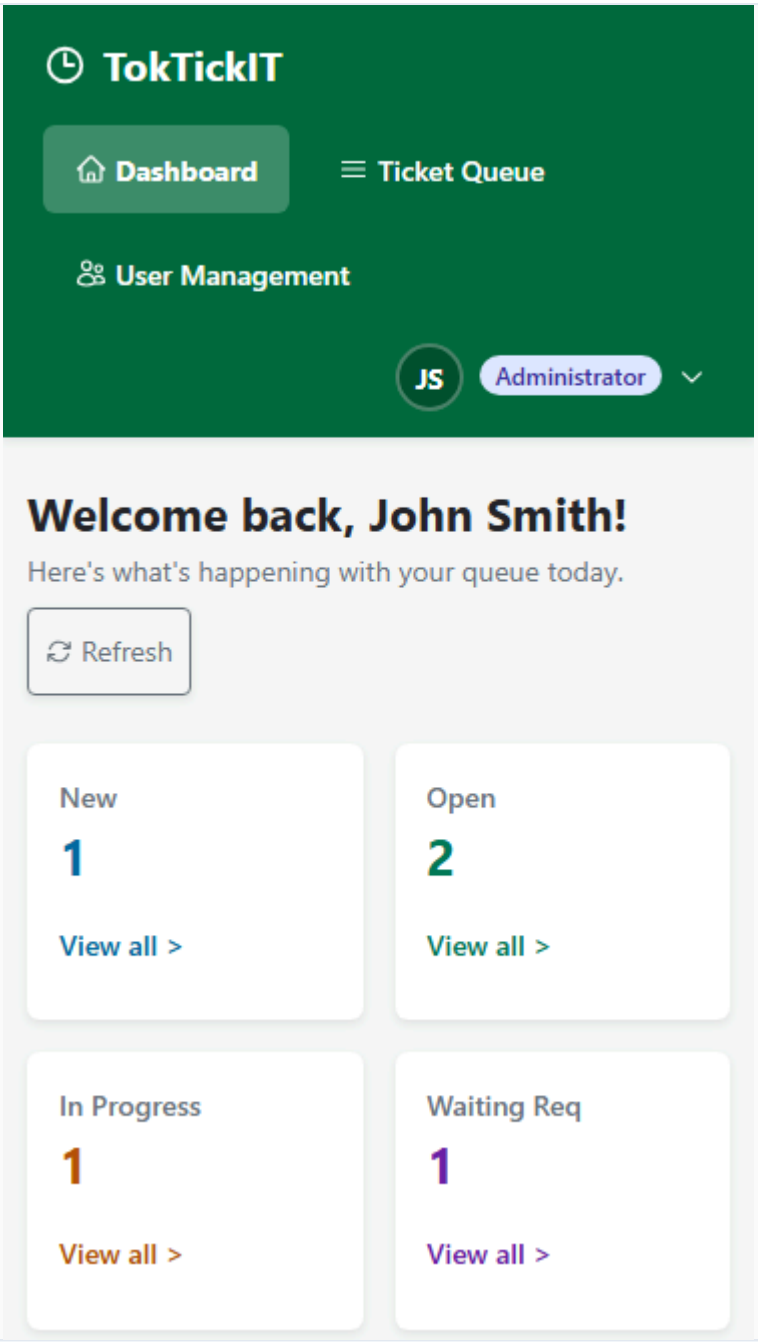


Figure 8.6: Administrator Dashboard on Mobile viewport.

3. Full Regression Suite across Labs 1–3 (E2E-10)

Automated test suite `e2e/lab-04/regression.spec.ts` executes 12 multi-viewport regression scenarios covering Lab 1 (health check, categories), Lab 2 (requester creation, My Tickets, detail, attachments), and Lab 3 (auth, forced password change, queue claiming, comments, notes, admin safety) with 100% pass rate.

Answer Part 9: Zen Green UI, Responsive, Accessibility & Final Polish

UI Specification Link: `docs/lab-04/ui-spec.md`

Design Tokens & Invariant Compliance

- **Zen Green Brand Tokens:** Primary `#006B3C`, Hover `#0B7A46`, Tint `#EAF6EF`, Canvas `#F5F7F6`, White `#FFFFFF`.
- **Amber Follow-up & Notes:** Border `#D97706`, Surface `#FEF3C7`, Text `#92400E`.
- **Editable vs. Readonly:** White inputs (`#FFFFFF`) for active forms; soft gray-green (`#F0F4F2`) for read-only displays.

Completed Responsive Visual Checklist

Requirement	Inspection Standard	Verification Method	Compliance Status
Zero Horizontal Overflow (RESP-04)	`scrollWidth ≤ innerWidth` across all devices	Automated `assertNoHorizontalOverflow()`	PASS
Minimum Touch Targets (RESP-03)	Buttons/inputs ≥ 44 × 44px on tablet/mobile	Automated `assertMinimumTouchTargets()`	PASS
Brand Color Tokens (VIS-01)	Strict adherence to #006B3C Zen Green	Automated CSS token assertions	PASS
Badge Semantic Contrast (VIS-02)	WCAG AA contrast for Role, Status, and Priority	Automated style tests in `ui-style.test.tsx`	PASS
Visible Focus Rings (VIS-03)	Visible #0B7A46 focus outline on Tab navigation	Sequential Tab traversal in Playwright & RTL	PASS
Zero Text Clipping (VIS-04)	Natural word wrapping without truncation	E2E responsive multi-device assertions	PASS
Form State Preservation (BR-18)	Preserves inputs upon error or 409 conflict	Component tests in `ActionsTakenSection.test.tsx`	PASS
Double-Click Protection (BR-19)	Disables submit button and shows spinner in-flight	E2E and component race condition tests	PASS

★ Visual QA & Responsive Invariants Checklist (VIS-01..04, RESP-01..04, BR-18..19)

TokTickIT Sprint 4 Verified

Invariant Code	Visual & Responsive Requirement	Verification Method	Result
RESP-01	Desktop layout at 1280×800 with full tables, cards, and multi-column navigation	Playwright Desktop viewport + Visual QA	PASS
RESP-02	Tablet layout at 768×1024 with 2-column responsive grid and modal auto-centering	Playwright Tablet viewport (AC-23)	PASS
RESP-03	Mobile layout at 375×667 with stacked cards and touch targets ≥ 44 × 44px	assertMinimumTouchTargets(page)	PASS
RESP-04	Zero horizontal overflow across all viewports (scrollWidth ≤ innerWidth)	assertNoHorizontalOverflow(page)	PASS
VIS-01	Zen Green brand token adherence (#006B3C, #0B7A46, #EAF6EF, #FFFFFF)	Computed CSS token assertions in RTL & Vitest	PASS
VIS-02	Distinct Amber theme for Follow-up Required (#D97706, #FEF3C7) and Internal Notes	Automated style assertions in ui-style.test.tsx	PASS
VIS-03	High-contrast visible focus outline (#0B7A46, 2px) on sequential Tab key traversal	Sequential Tab key traversal in E2E & RTL	PASS
VIS-04	Zero text clipping, word wrapping on long text, and responsive metric cards	E2E responsive multi-device assertions	PASS

Figure 9.1: Visual QA and responsive design invariant verification matrix.

Complete 24 Screenshot Evidence Inventory

All 24 screenshot evidence artifacts captured across Desktop (1280×800), Tablet (768×1024), and Mobile (375×667) are committed and verified under `artifacts/lab-04/screenshots/`.